

The Wedge Method

Let the frame be lumpy. Get the skin and the floor right. A dome from split logs, and the two layers that decide whether it lasts

Built 2026-09-25. Every figure in this book was computed at build time from the geometry in the DomeSim project.

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PART 1

What Actually Has to Be Right

“A dome is a frame, a skin and a floor -- and the frame is the part you are allowed to get wrong.”

Three chapters that invert the usual advice, and one practical consequence: build the frame loose and the two layers that matter stay changeable for the life of the building.

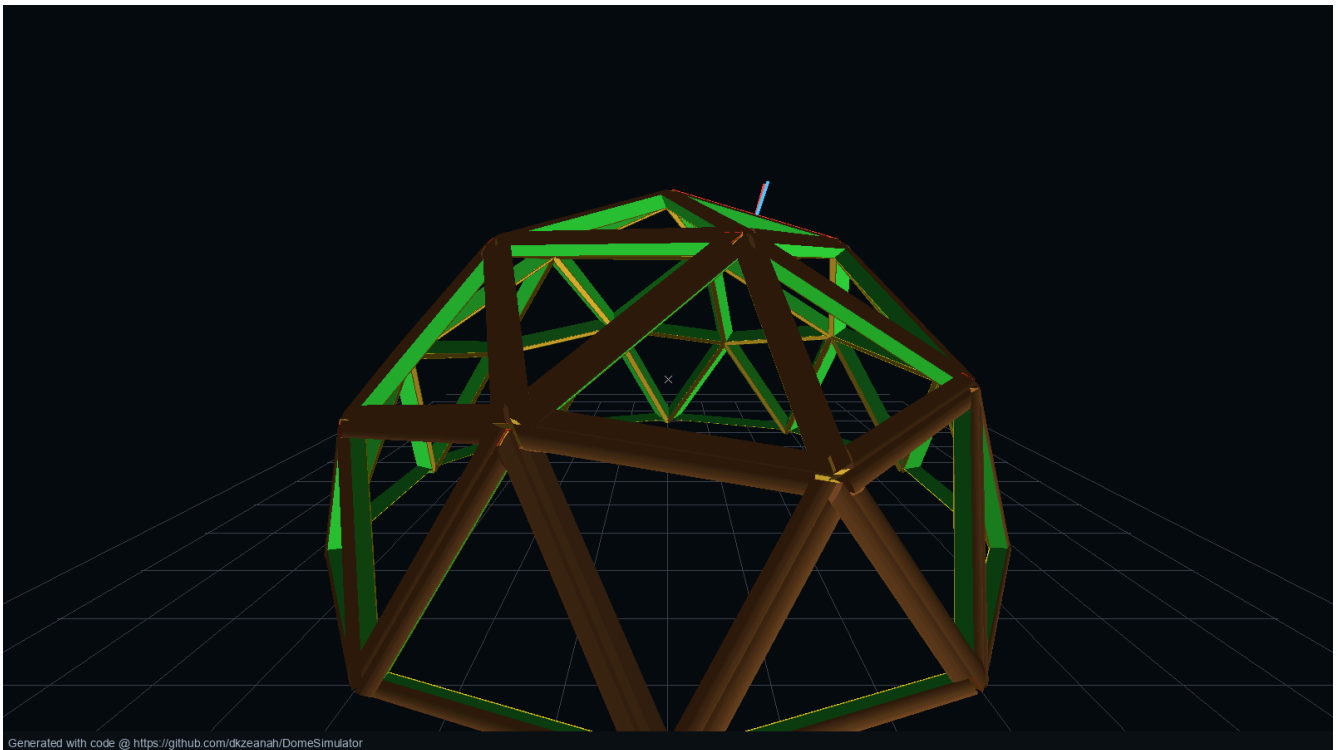
Your Skull Is Not Symmetrical Either

The frame is the part you are allowed to get wrong, and almost every dome book has this backwards

Every dome book I have read opens by telling you to be precise.

Measure twice. Hold your angles. A geodesic dome is a mathematical object and you are about to approximate one out of wood, so the closer you get to the maths the better your building will be.

I believed that for about a year, and it cost me most of a year.



The reference build. Nothing about it is as accurate as it looks.

The afternoon I stopped believing it

I had a frame up. Not this one — an earlier one, made of whatever the chainsaw had produced that week, with no two sticks alike. I had built it to prove to myself that the method could not work, so that I could stop thinking about it and go and buy some lumber like a normal person.

It stood. It stood with a stick in it that was out by most of an inch, and a vertex where three members met that I could put my whole thumb into.

I pushed it. I hung off it. I put a ladder against it and climbed on it, which I do not recommend and did anyway.

It did not care.

Why it did not care

A dome carries its loads *axially*. The members are in compression along their own length — pushing end to end, the way a stack of bricks pushes — and a triangulated shell passes a load from one triangle to the next all the way round, so no single member is ever being asked to do very much on its own.

Put a stick in slightly out of position and two things happen. The triangle it is in is very slightly a different triangle. And the load path shifts by that same very slight amount.

Neither of those is an error that grows. A frame is not a tower of tolerances stacking up toward a failure; it is a shell that absorbs the difference and carries on. The error goes into the seam — and there is already a key in the seam, because there was always going to be one.

The lumpy skull

You have an asymmetric skull.

Genuinely: measure it. One side of your head is further from the midline than the other, one eye socket is fractionally lower, your jaw is not centred. This is true of essentially everybody and it is why nobody's face looks right when you mirror it.

Nobody likes you any less for it, and your skull opposes a baseball bat exactly as well as a perfectly symmetrical one would — because the thing that stops the bat is the *shape*, a closed curved shell in compression, and not the precision.

That is a dome. The geometry does the work. The accuracy is decoration.

What this book does with that

It spends it.

If the frame does not need precision, then the frame does not need a mill. It does not need a planer, or a moisture-controlled shed, or lumber bought to a tolerance somebody else held for you. It can be made of logs split with a wedge, on the ground, in a field, by one person who is not very good at it yet.

That is the whole method. Everything else in this book — the 45.0-degree sector, the pinwheel joint, the one flat board that makes 40 identical panels — is a consequence of deciding that the frame is the part you are allowed to get wrong.

And what it does not spend

Here is the part that took me another year to learn, and it is the reason this book exists rather than being a pamphlet about splitting logs.

The permission is not general. It applies to the frame and it applies to nothing else.

The skin has to be right, because water does not negotiate and does not announce itself. The floor has to be right, because everything you own sits on it and because the ground sends its water up.

So the advice most dome books give is not merely unhelpful. It is aimed at the wrong layer. It asks for care where care is cheap to skip, and it treats as finishing work the two layers that decide whether the building is still there in thirty years.

The next two chapters are about those two layers. Then we can go and cut something.

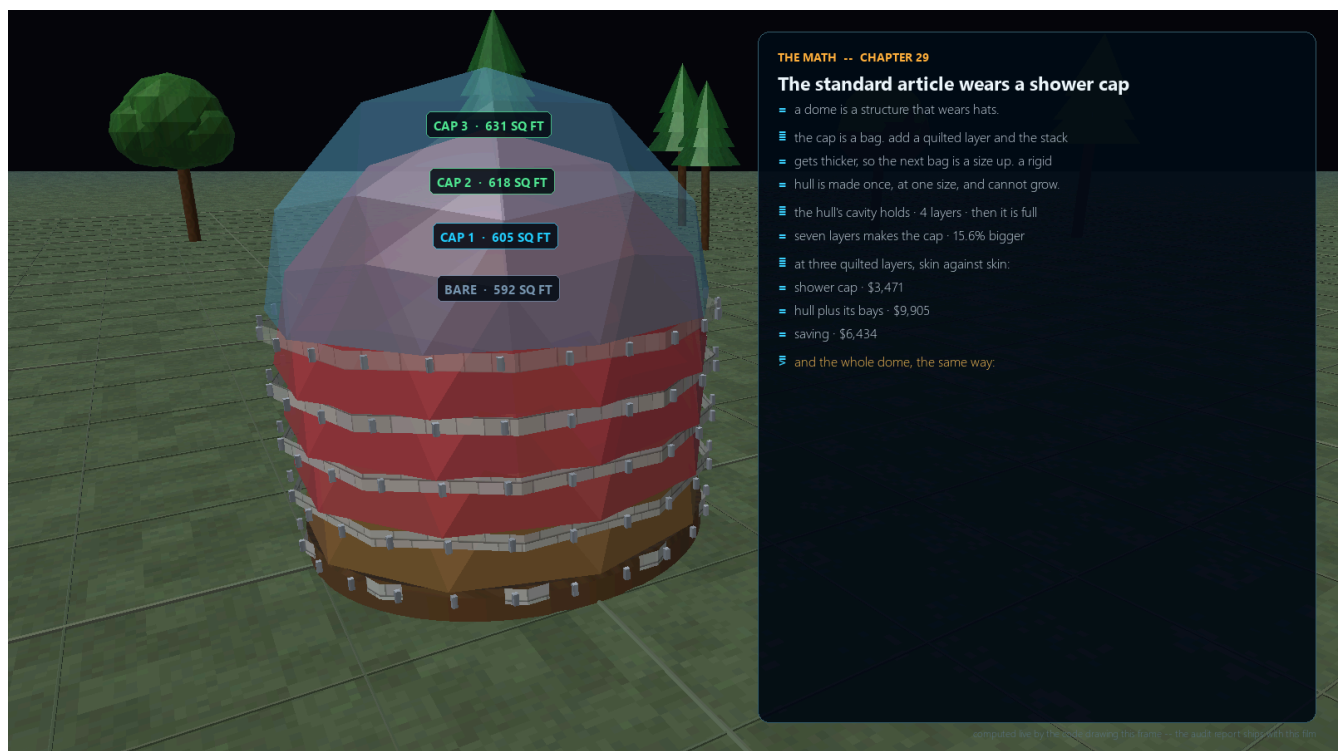
Water Is the Silent Killer

The skin is the part that has to be right, and it is the part every dome book treats as a finish

A frame that is out by an inch stands for fifty years.

A skin that leaks for one winter takes the frame with it.

That asymmetry is the most useful thing in this book, and almost every dome book has it the other way round.



One watertight layer over everything else.

Why water gets to decide

Structural failure is loud. Something cracks, something sags, something you can see from across the room. You get warning, and the warning arrives in a form that makes you do something about it.

Rot is not loud. Rot is a slow chemical argument between a fungus and your building, conducted at about twenty per cent moisture content, entirely out of sight, and it is usually two or three years old by the time anything shows on a surface you look at.

By then you are not repairing a leak. You are replacing members, and the members are the thing this whole method spent its cleverness on making cheap — so you are not even losing much money. You are losing the fortnight.

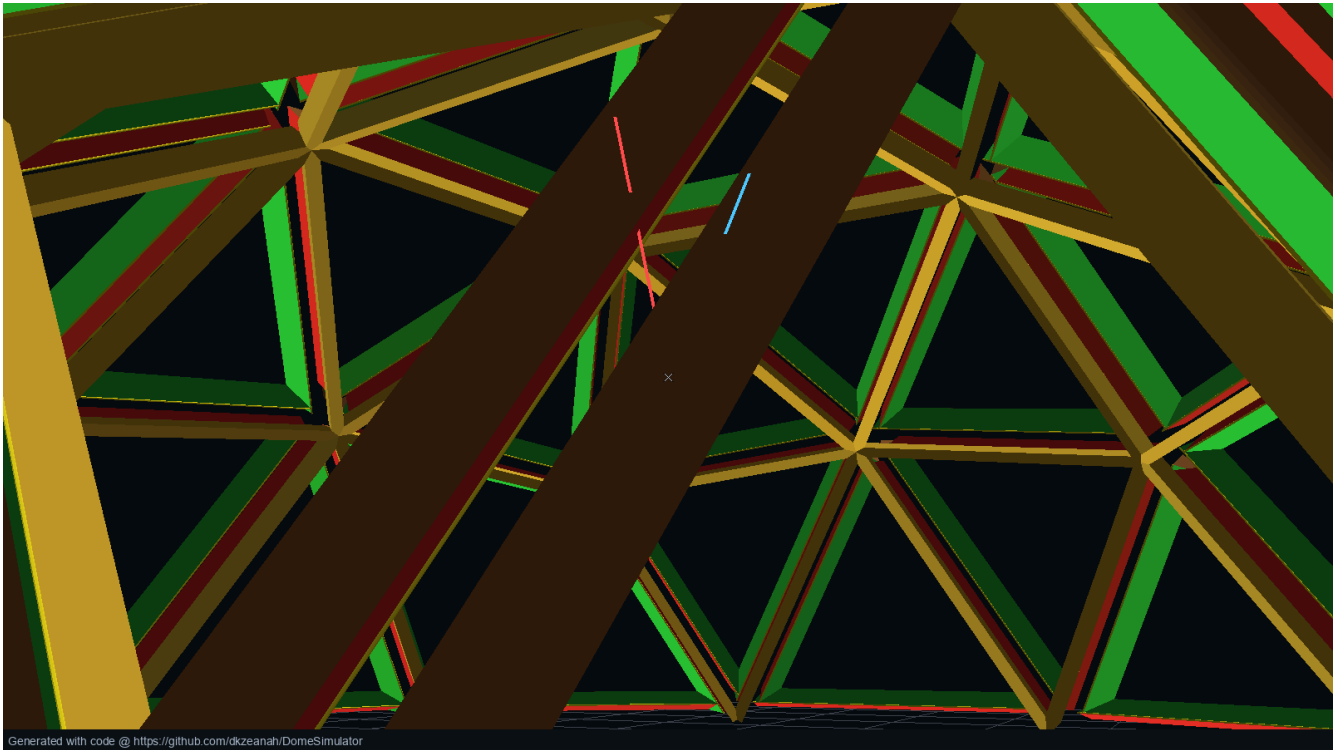
That is the trade nobody states plainly: the frame is cheap to build and expensive to replace, because replacing it means taking the building apart.

Where a dome actually leaks

Not through the panels. A panel is a large flat sheet with no holes in it and it is the easiest part of any building to keep water out of.

Domes leak at seams, at penetrations, and at the apex, in that order — and a 2V dome has a great many seams.

This one has 55 interior seams, which is 309 feet of joint. Every foot of that is a place where two panels meet and where somebody has to have done something deliberate.



Every seam in the dome, and the key in each one.

Three hundred feet of joint is not a detail. It is the single largest risk surface the building has, and it is spread out so evenly that no one part of it looks important enough to worry about.

The move that removes the category

You do not solve three hundred feet of joint by making each joint better.

You solve it by putting one continuous layer over all of it, so that the joints stop being the outside of the building.

That is what the shower cap is. One watertight membrane over the whole shell, strapped down, with nothing structural fastened through it and no seam in it that is not a welded one. The panels underneath do not have to be watertight. The seams underneath do not have to be watertight. They have to be dry, and they are dry because something above them is doing that job and only that job.

The chapter on the cap (11) has the physics and the arithmetic. What matters here is the ordering: **one layer whose only job is water, and it goes on last and comes off first.**

And the floor

The floor is the second thing, and for a different reason.

It is the surface you are in contact with all day. It carries everything you own. It is the only part of the building that is in permanent contact with the ground, which means it is where ground moisture arrives, and unlike a wall it cannot dry to the outside because there is no outside — there is dirt.

A floor that is cold, or damp, or that flexes, is a floor you notice every hour you are awake. A frame that is out of true is a frame you never think about again after the first week.

Get the floor right. Chapter 13 says what right means, and what the five ways of doing it cost on this footprint.

The hierarchy, stated once

1. **The skin.** Water. One continuous layer, no compromises, no penetrations you did not design.
2. **The floor.** Comfort, dryness, and everything you own.
3. **The frame.** Standing up, which it will do whether or not you were careful.

If you read nothing else in this book, invert your instincts in that order. It is worth more than every angle in the appendix.

The catch, and the next chapter

There is an obvious objection, and it is a good one.

The skin and the floor are the expensive parts. They are the parts you cannot cut out of a tree with a chainsaw. Telling somebody that the two layers that matter are also the two layers that cost money is not obviously helpful.

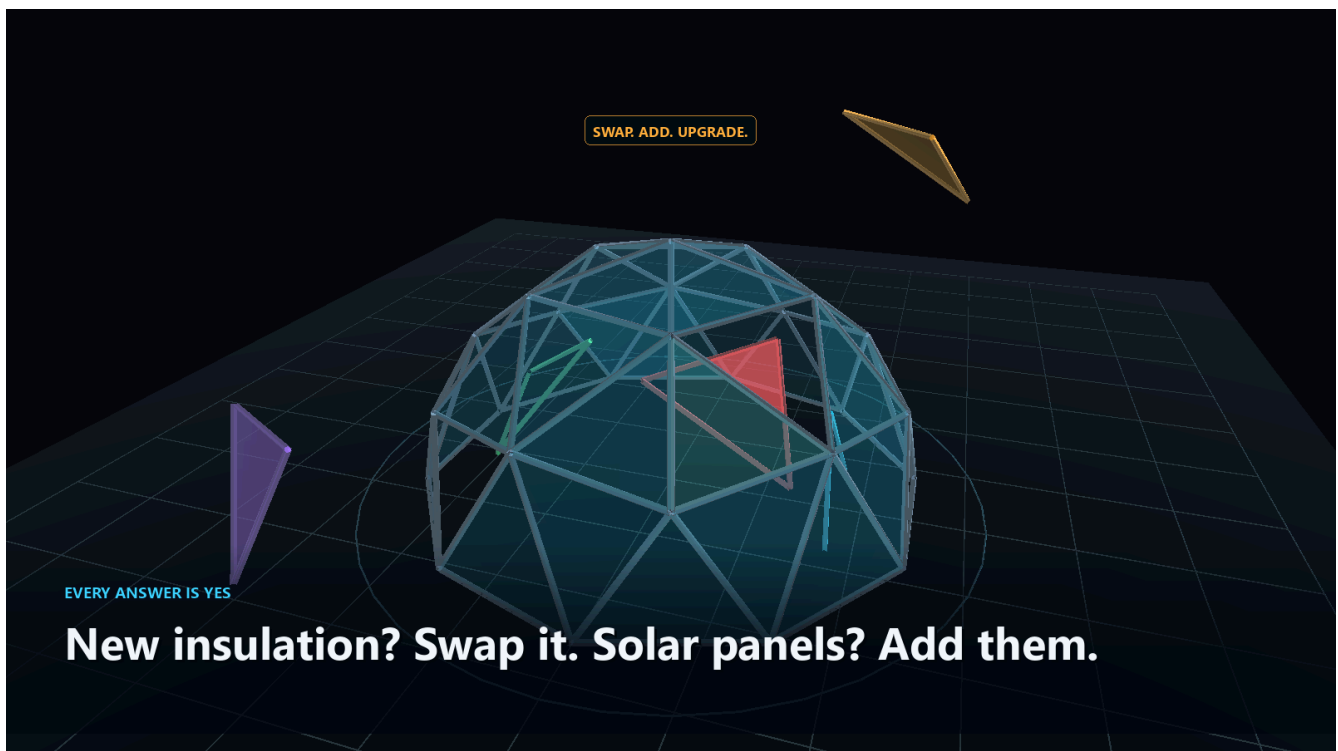
The answer is that they are also the two layers you can *defer* — and the next chapter is about the four things you do while building the frame that keep that option open for the life of the building.

Build It So You Can Change Your Mind

The skin and the floor are the two things that matter and the two things you can defer -- if the frame is built for it

The two layers that matter most are the two layers you can put off.

That sounds like a consolation and it is actually the design. A frame is a commitment: 120 members, cut to two lengths, and once the first panel is closed the shape of the building is decided. A skin is not. A floor is not. Those are things the building *wears*, and a building can change what it wears.



New insulation? Swap it. Solar panels? Add them.

The condition

It only works if you build for it, and the decisions are all made in the first week, before there is a building to look at.

None of them costs anything meaningful at the time. All of them are effectively impossible to add later.

The four things

Threaded inserts in the wedge faces, not screws into end grain.

A panel that will be taken off and put back needs somewhere to bolt to, and end grain is not it — the second time you drive a screw into the same hole it is holding about half of what it held the first time, and the fourth time it is holding nothing. An insert set into the flat sawn face of a wedge is a machine thread in a piece of metal. It does not care how many times you use it.

This is the single highest-leverage decision in the build, and it happens while the panel is still flat on the jig.

A cap sized a layer larger than it needs.

The outer skin goes over whatever is under it, so its size depends on how much is under it. A cap cut to fit a bare shell will not go over a shell wearing two quilted layers, and you find that out on the day you want to add the second layer, in November.

Chapter 12 has the arithmetic — each hat is a size up, and the sequence is not one number. Buy one size up from what you need on day one.

A floor that clamps rather than is built in.

A floor framed into the base ring is part of the structure. A floor that is a hub, radial spokes and decking, clamped to something in the middle, is a component — it can come out, go back, be replaced with a better one, or be left behind.

It also turns out to be the thing that makes a floating dome possible at all, which was not the reason for doing it.

Services in the seam channel, not in the wall.

The seam leaves a gap that has to be filled with something. Fill it with a hose and the building's plumbing is in its structure, reachable from outside by taking off one panel, and no pipe has ever been

buried in anything.

An A seam's channel takes a line up to 2.878 inches across. That is not a tight fit for the things a small building actually needs.

What deferring costs

Honestly: some money, later, in exchange for less money now.

The standard article in this book's cost model lists at \$12,036, built to \$10,030. The shell inside that is \$3,265. The bare version of the same cap — no quilted layers at all — is \$3,265, and each layer you add afterwards is a few tens of dollars of material and an evening.

So the deferral is real and it is cheap. What you are buying with the four decisions above is the right to spend that money in the order your life actually allows, rather than in the order a building would prefer.

The failure this prevents

I have watched somebody build a beautiful frame, run out of money, put a tarpaulin over it for a winter, and lose the frame.

Not because the tarpaulin was a bad idea. A tarpaulin is a perfectly good temporary skin. They lost it because there was nowhere for a real skin to *attach* when the money came back — no inserts, no strap points, nothing but a frame that had been designed on the assumption that the skin would be figured out later by somebody with more time.

Later is not a plan. Later is a thing you design for.

Where this leaves us

Three chapters in, the argument of this book is complete:

- the frame is allowed to be messy, which is what lets a split log be a structural member;
- the skin and the floor are not, because water and comfort do not forgive;
- and the two layers that are not allowed to be messy are the two you can postpone, upgrade and swap — if you spend four decisions on it in week one.

The rest of the book is how. It starts where the material does, with a tree.

PART 2

The Stick

“Every radius of a round log is the same as every other. That is the whole trick, and the tree held the tolerance.”

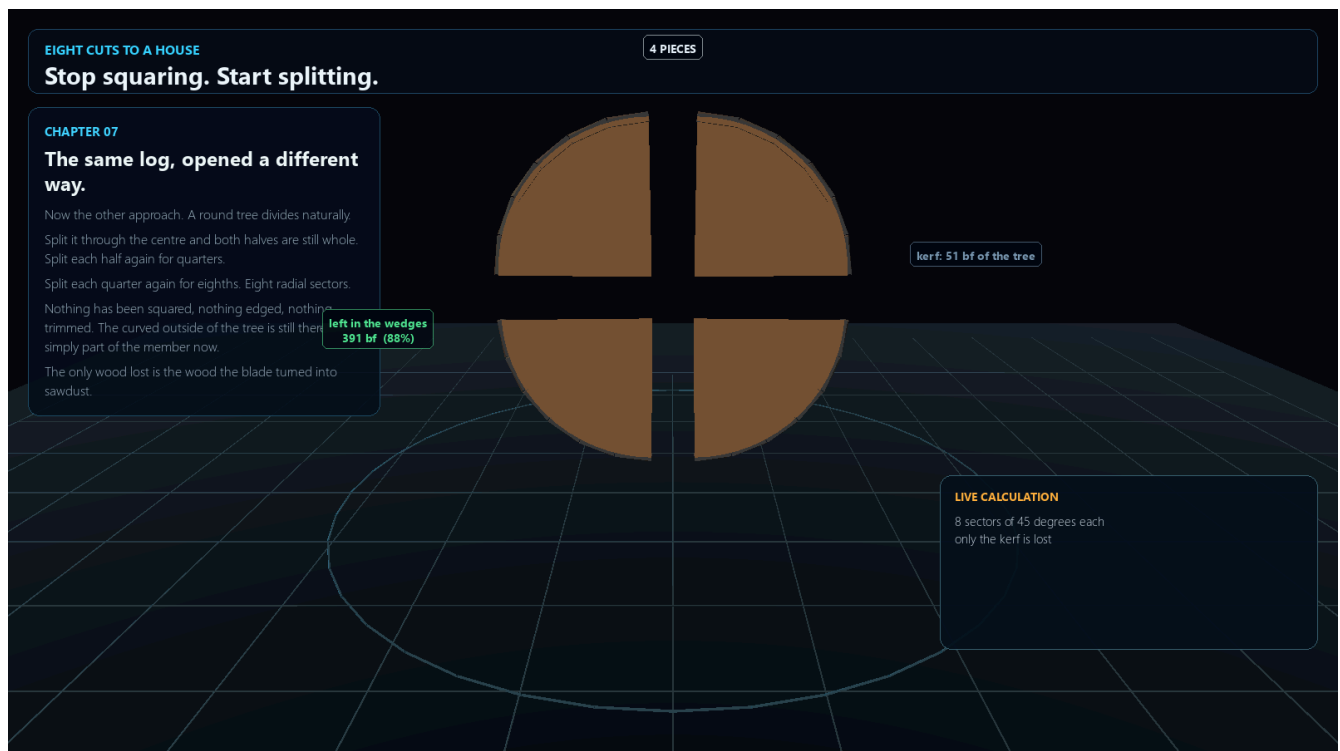
How a standing tree becomes 120 structural members, what a 45-degree sector is, and the two lengths you actually cut.

Stop Squaring. Start Splitting.

A round log is already rotationally symmetric, and three splits give you eight identical sticks nobody had to measure

I was sitting on a log when this stopped being difficult.

Not thinking about it. Sitting on it, having spent the morning failing to make a connector that would join any stick to any other stick at any angle, which is a problem you can work on forever because it has no answer.



The same log, opened a different way.

The wrong question

The question I had been asking was: *how do I join sticks that are all different?*

Everybody's answer to that is the same. Mill them. Square them up, run them through a planer, and now they are all the same because you made them the same.

That answer costs a mill, a planer, a shed to keep the mill dry, the fuel to run it, and — this is the part nobody mentions — about half the tree. A round log does not contain a rectangle. Squaring one is the act of cutting a rectangle out of a circle and calling the rest waste.

The better question

Sitting on the log, I noticed the log.

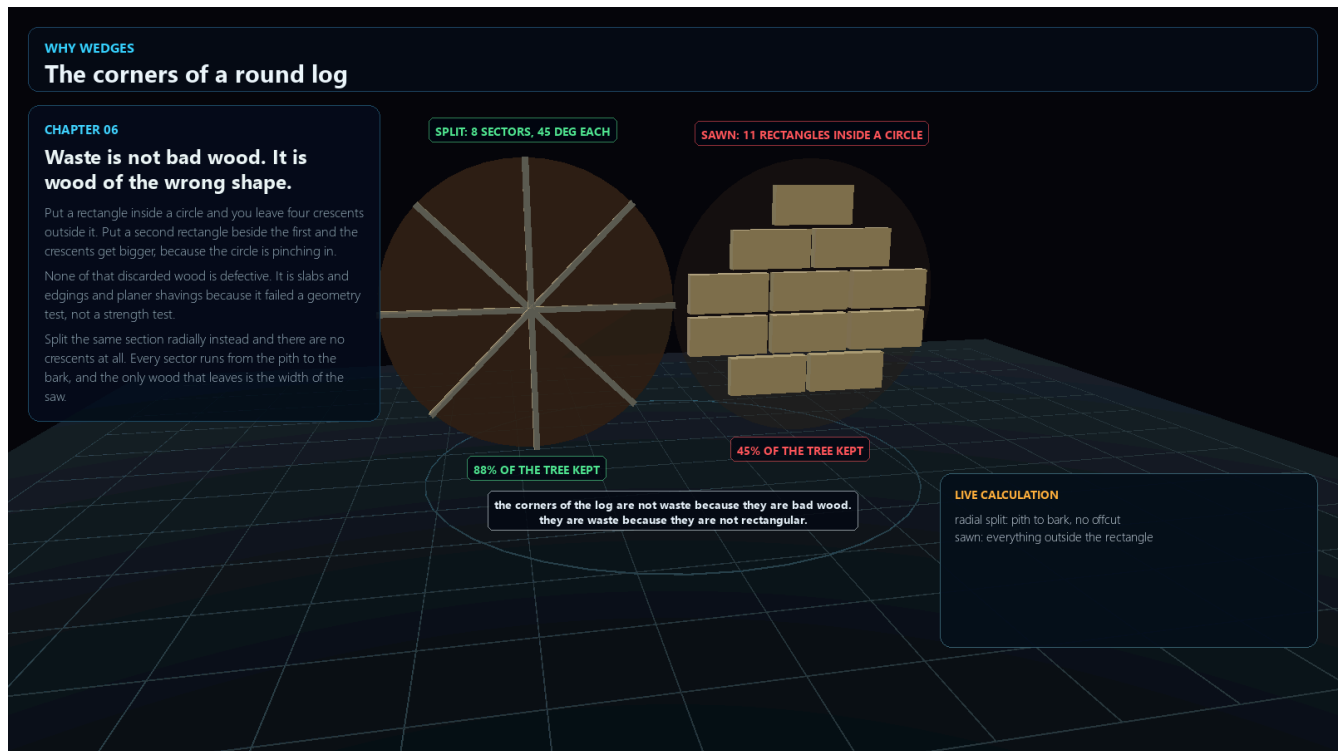
A round trunk is rotationally symmetric. Every radius is the same as every other radius. That is not a thing you achieve; it is a thing the tree did while you were not there.

So split it. Split it through the middle and you get two identical halves — identical not because you were careful but because the log was round before you touched it. Split those and you get four. Split those and you get 8.

Three passes. Eight sticks. Every one of them a 45.0-degree sector of the same circle, to whatever tolerance the tree was round to, which is a tolerance nobody had to hold and nobody had to check.

That was the afternoon. It is not a clever idea and it took me a year to have it, because I was busy asking a question that had no answer.

What the mill is actually for



The corners of a round log are not bad wood. They are wood of the wrong shape.

Here is the thing to be careful about, and I got it wrong in an earlier version of this argument.

The corners a mill throws away are not bad wood. They are perfectly good wood of the wrong *shape*. A sawmill exists to turn trees into a standard rectangular product that can be stacked, graded, priced, shipped and sold by people who will never see the tree. It is extremely good at that, and the waste is the price of the standardisation, not of incompetence.

Splitting does not beat sawing at making lumber. It beats sawing at making *this* — a structural member for a frame that does not care about rectangles.

That distinction matters because it tells you when the method stops working. Build something with right angles in it and you want a mill. Build a triangulated shell and you do not.

What comes off the log

At the reference build's 12.0-inch trunk, each sector is 4.592 inches across its bark face and 6.0 inches deep from point to bark. The frame needs 120 of them, which is 664 feet of stick.

Nothing in that sentence required a tape measure on the tree. It required a froe, a mallet, and a willingness to let the log decide the section.

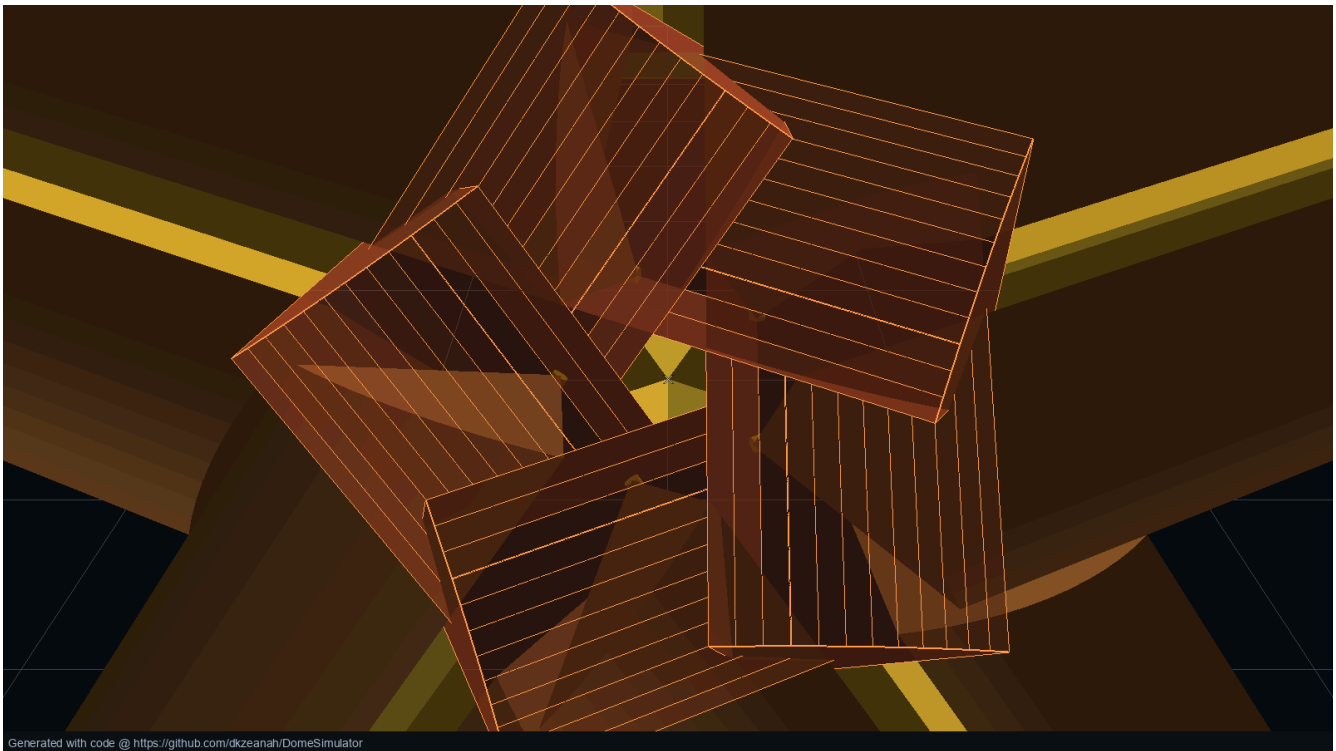
The next chapter is what that section actually is, and which way round to put it — which is the first decision in this build you cannot take back.

One Eighth of a Tree, Used as a Stick

What a 45-degree sector actually is, and why its shape is an advantage rather than something to apologise for

A member in this frame is a 45.0-degree slice of a 12.0-inch log. It has a point, two flat sawn faces, and a curved bark face.

That is the whole part. Nothing is machined.



The end of a member: two sawn faces and a bark face, and nothing machined.

What the shape gives you for free

Two reference surfaces. The split faces are flat and they are at a known angle to each other — 45.0 degrees, because that is what a 8-way split means. Every jig in this book locates off those two faces.

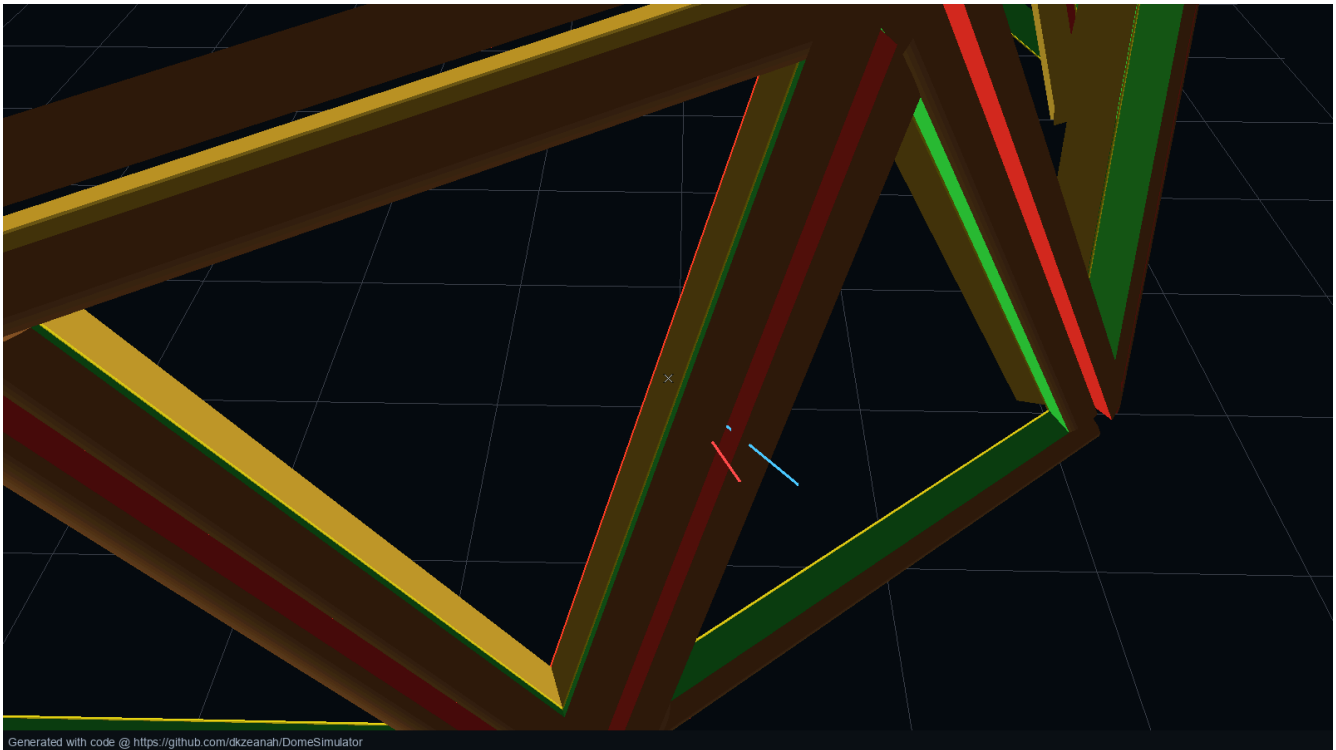
You never reference the bark, which is lumpy, and you never reference a measurement, which is a thing that can be wrong.

A point that tells you which way round it is. A rectangular stick has a four-fold ambiguity: drop it in rotated ninety degrees and it still looks fine. A sector has one obvious orientation, and — as Chapter 9 shows — the fixture is built so a stick dropped in the wrong way round physically does not seat.

Depth where you want it. The section is deep from point to bark (6.0 inches here) and narrower across the face (4.592 inches). Stand it point-inward and the depth is in the direction the shell wants stiffness.

Which way the point faces

This is the first decision in the build you cannot take back, because it changes the seam, the key, and the shape of every panel.



Point inward: the reference build.

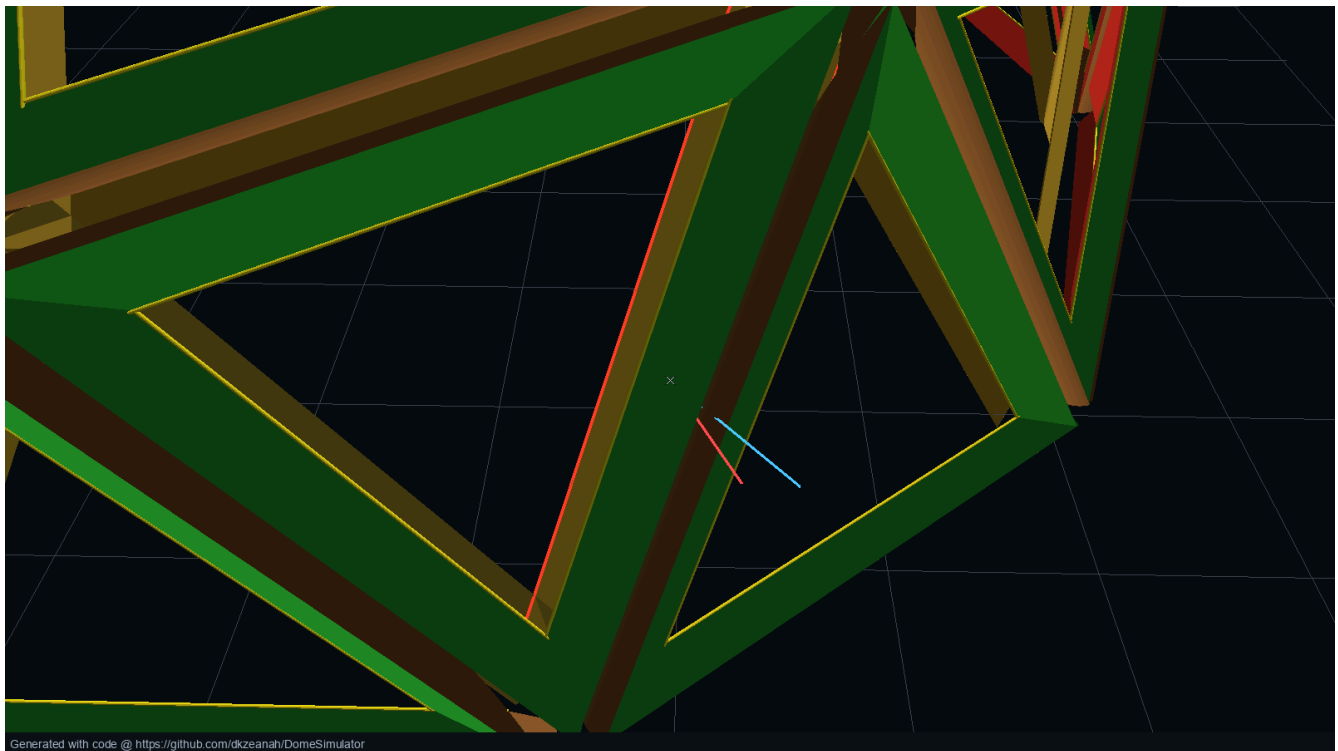
The reference build faces the point **into the dome**. The bark faces are outside, the flat sawn faces look at each other across every seam, and the leftover sector angle opens toward the inside.

The consequences, from the solve: an A seam folds 18.029 degrees and leaves 26.971 degrees of sector angle over; a B seam folds 22.459 and leaves 22.541.

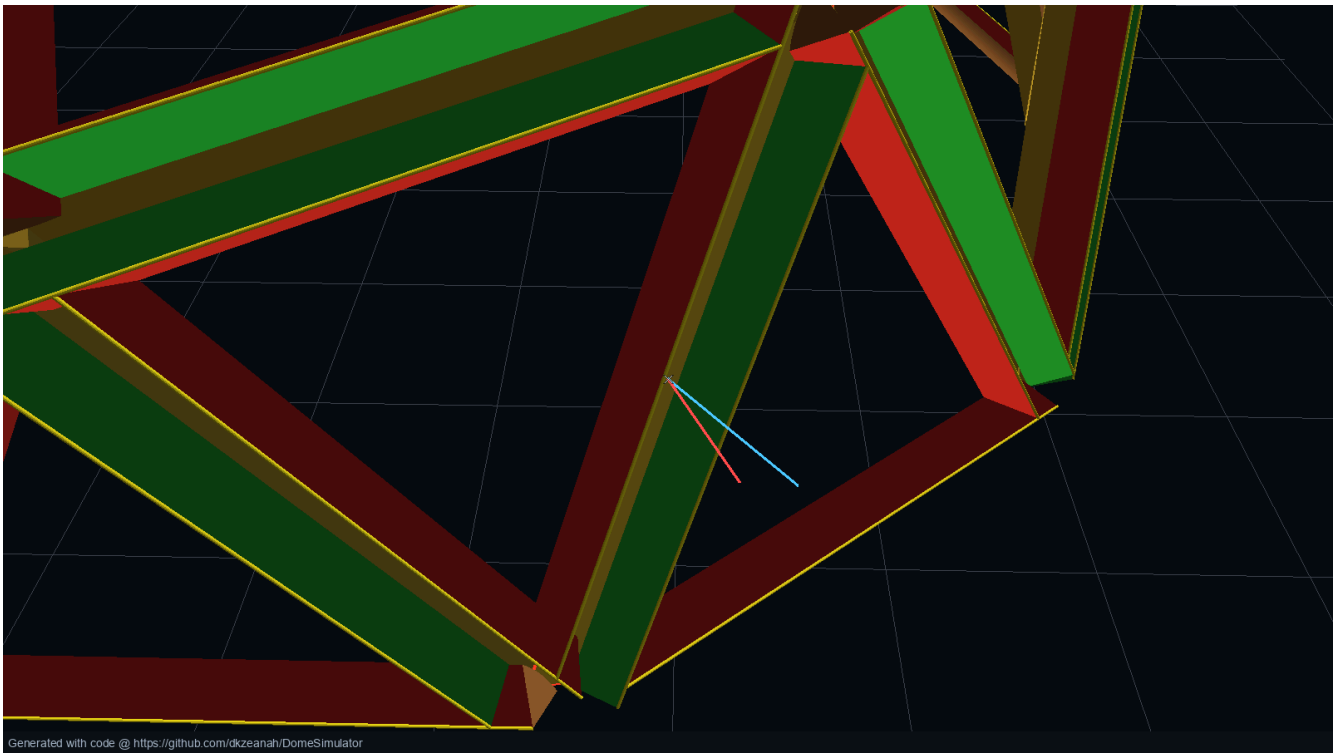
Each pair sums to 45.0. That is not a coincidence and it is the closest thing this method has to a theorem: **the sector you split out of the tree is exactly the seam budget the dome needs.**

The other three

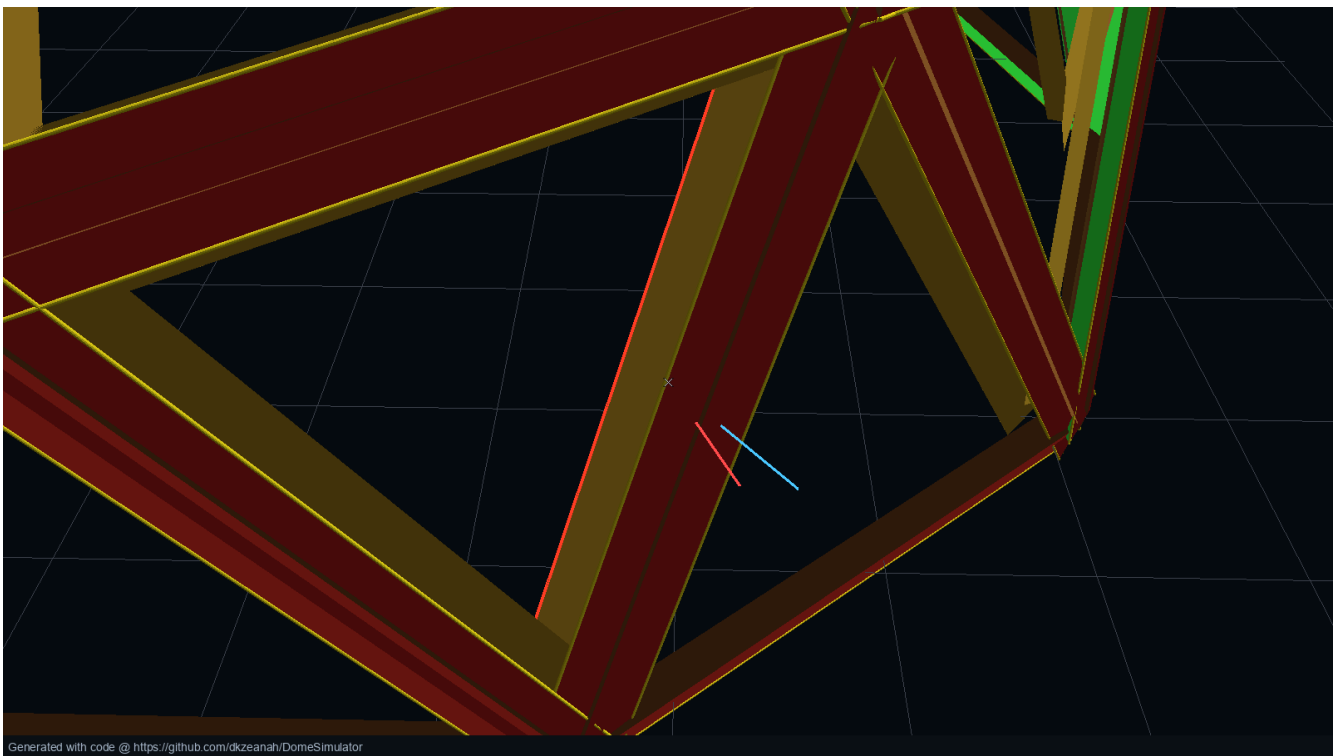
The solver will build all four. They are not equally good and they are not equally bad, and the book shows them because a reader who wants something different should be able to see what they are choosing.



Points apart, each into its own triangle.



Both points outward, at the sky.



Points toward each other, across the seam.

Point outward puts the flat faces where the weather is, which is a real argument for it: a flat outer face is easier to seal a panel against than a curved one. It costs you a key more than twice the size, and it puts the narrow point where the shell wants depth.

Points apart, and points across, both produce a seam that wants a key around eleven and a half inches across its base. That is not a key. That is a beam you are now also making.

Why the bark stays on

It is free, it is already there, and taking it off is an operation on 120 sticks.

The honest version: bark is not a finish and it will come off in its own time on the outside faces, in patches, over years. On a frame that is inside a weatherproof skin — which is the argument of Chapter 2 — that is a cosmetic event rather than a structural one.

If you want it off, take it off green, with a drawknife, in the first week when it is still wet and comes away in strips. Do not plan to do it later.

The one number to hold on to

A member's section is set by the log, and the log is set by what you have. Everything downstream — the seam key's size, the depth of the wall, how many trees you need — follows from the trunk diameter and the split count, and Chapter 16 prices what happens when you change either.

What does *not* change is the two lengths you cut them to, and that is the next chapter.

Two Lengths, and Neither Is the Chord

72.000 and 63.670 inches are what the geometry says; what you cut is those minus a bite that does not scale

A 2V dome has two member lengths. This is the page you come back to.

The two chords

A chord is the straight-line distance between two vertices of the sphere. It is the number every geodesic reference publishes, and it is a property of the shape rather than of your building:

A chord = 0.618034 x radius

B chord = 0.546533 x radius

At this book's radius of 116.4984 inches, that is 72.000 and 63.670 inches.

The A chord being a round 72 is not luck. The dome was sized from the stick: a six-foot member is what comes comfortably out of a twelve-foot log section, what one person carries, and what fits in a pickup. The diameter is the answer to the member, not the other way round.

The chord is not what you cut

Neither end of a member reaches the vertex it is named after.

One end butts into the *side* of its neighbour, short of the corner. The other runs past the mathematical vertex so that the previous member can butt into its side. Chapter 7 is why. The consequence is that the joint takes a bite out of each end, and the cut length is:

cut = chord - bite

For this frame:

A members	72.000	-	1.818	=	70.182 in
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B members	63.670	-	0.990	=	62.681 in
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The bite does not scale

This is the part that catches people, and it caught this book.

The bite is set by **how wide the member is**. A member does not get wider because the dome gets bigger — you are still splitting the same size of log — so the bite is the same number at every diameter.

Solve the same pinwheel at a four-foot radius and at a fifteen-foot one with the same stick and the bite comes out identical to four decimal places.

So: **scale the chord, subtract the bite**. Not chord times a ratio.

If you take one thing from this chapter, take that.

The bite used here

A members	1.818 in
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B members	0.990 in
-----------	----------

Both at this book's 4.592-inch member. A wider member takes a bigger bite, roughly in proportion, so a frame cut from heavier stock wants its own solve rather than this number.

What this book got wrong

An earlier version of the reference-design tables scaled the bite with the dome.

It is correct at the reference build — that is how it got past everybody — and wrong at every other size. At eight feet across it made every stick more than an inch too long. At twenty-four feet it made them half an inch short. Every stick, all 120 of them, in the two sizes a first-time builder is most likely to pick.

It was found by a program that reads the finished book and checks every number in it against what the code can produce. That program is in Chapter 18, and this is the error it was written to catch.

Two models, and they disagree

This repository solves the pinwheel twice.

The raw-wedge solver treats a member as a real 45.0-degree log sector. That is where the bites above come from, it is what the simulator draws, and it is what the reference build is cut to.

A second, simpler model treats a member as a rectangular band lying with its face on the edge line. It gives a bite of 9.253 inches for the same stick — several times larger.

Neither is a mistake. They are answers about differently shaped sticks.

The numbers in this book are the first. **If you are cutting rectangular stock rather than log sectors, this table does not describe your frame**, and the difference is inches on every member.

The row to check yourself against

19.42 feet across: A chord 72.000, A cut 70.182.

If your own arithmetic gives that, the rest will behave.

PART 3

The Panel

“Forty panels come off one board, and nothing on the board is adjusted between them. Panel forty is panel one.”

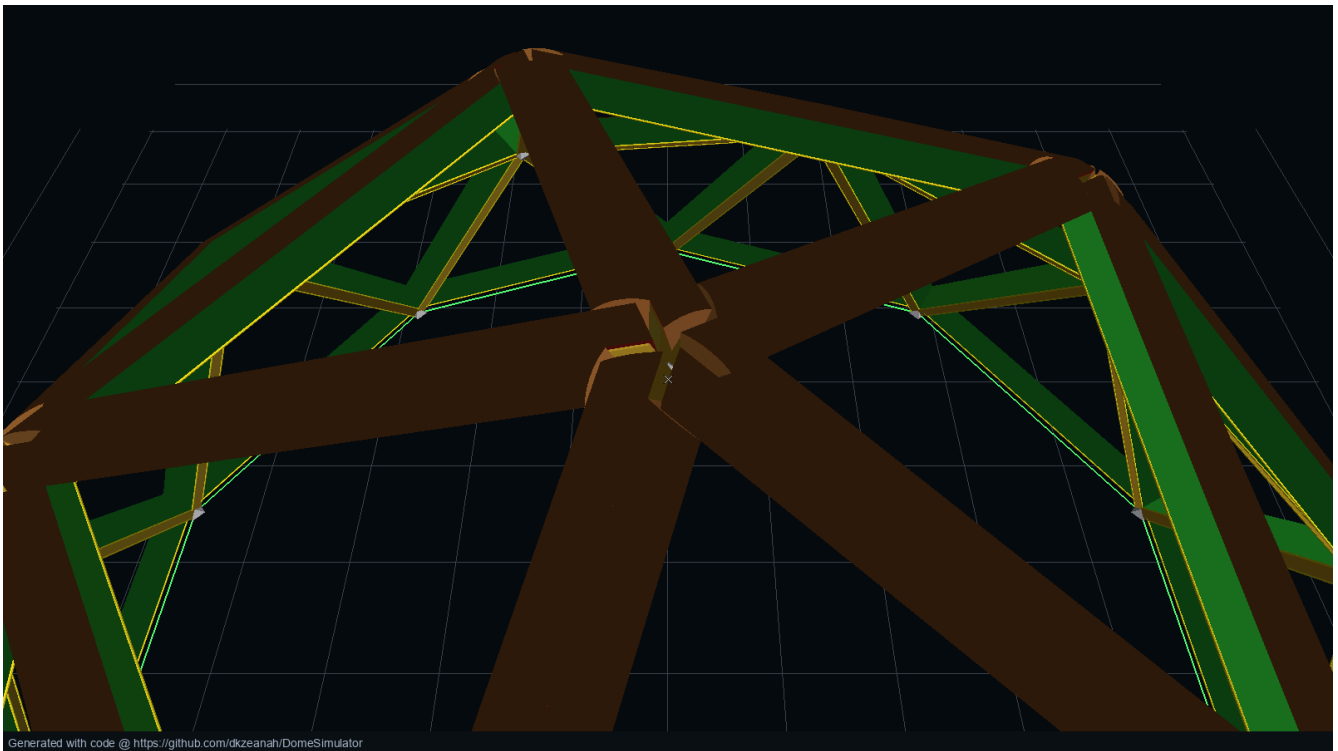
The joint that removes the hub, the nine processes that make the whole shop, and the one fixture that makes forty panels the same.

Nothing Meets at the Corner

Three sticks, each butting into the side of the next, and not one of them reaching the vertex it is named after

In a classical geodesic frame, five or six struts arrive at a point and something has to hold them there. That something is a hub, and hubs are why most people give up on domes.

This frame has no hubs, because nothing arrives at the point.



Five triangles meeting, and an empty corner.

The pinwheel

Each panel is built complete before it goes anywhere near its neighbours: three members, closed into their own triangle.

Inside that triangle, each member's end butts into the **side** of the next one, a little way along from the corner. Not into its end. Into its face. The three of them chase each other round the triangle, which is why it is called a pinwheel, and the mathematical vertex in the middle of that arrangement has nothing in it at all.

Three consequences, and they are the argument for the whole method:

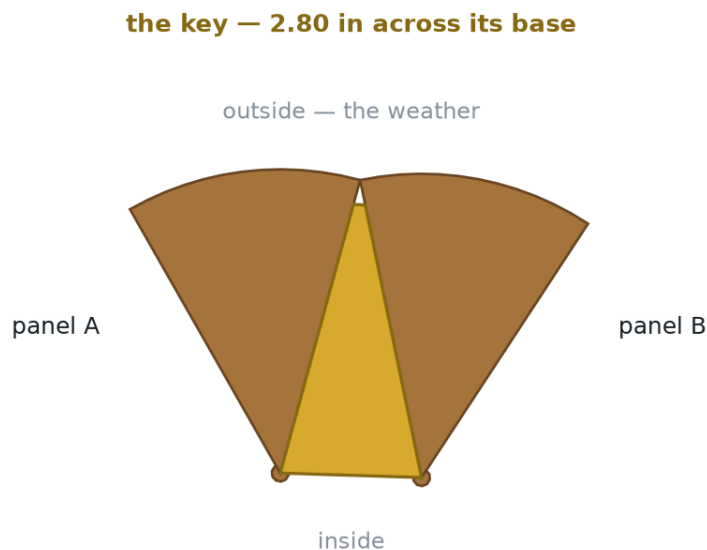
No hub. There is nothing to buy, machine or get wrong.

Flat sawn face on flat sawn face. A split wedge already has two flat faces at a known angle. The joint presses one against another. Nothing is mitred to meet at a point.

The corner is empty, so the corner cannot be crowded. The failure mode of a hubbed dome — five struts whose ends all want the same cubic inch — does not exist here.

What is actually at a seam

Where two panels meet there are **two** members, one belonging to each, with the key between them.



A seam · the two sawn faces are 26.97° apart · fold 18.03° · 45° sector of a 12 in log

A seam in cross-section, at the angle the solver puts it: two sectors, the key between them, drawn to the solved dihedral.

That picture is a real cut. A plane through the seam, perpendicular to its own run, intersected with the solver's own geometry — not a diagram of the idea. Two 45.0-degree sectors, points inward, and the 26.971 degrees of sector angle they cannot close between them.

The key fills that. It is 2.798 inches across its base on an A seam and 2.345 on a B, and it is not a manufactured part: it is the truncated point of another raw sector, cut from the same log at the same angle.

The price of a panel being a thing

40 panels times three members is 120. The shell has only 65 edges.

So 55 members exist twice over, once for each panel meeting there. That is nearly twice the timber a shared-strut frame would use.

It is not lumber somebody forgot to remove. It is the price of a panel being an **object** rather than a position in an assembly — and what you buy with it is everything in Chapter 9: a panel you can build flat, check flat, carry, sheathe and lift as a finished thing, on a bench, before anything is standing.

Two trees instead of one and a half, against forty panels you can make indoors in winter. That has been an easy trade every time I have had to make it.

Where the error goes

Chapter 1 claimed the frame absorbs error. This is the mechanism.

A member that is a bit long, a bit short, or at a bit of an angle changes the width of the gap at its seam. The gap already has a key in it. A key is a wedge, and a wedge accommodates a range of gaps by sitting a little deeper or a little shallower.

So the error does not accumulate around the triangle and it does not propagate into the neighbours. It lands in a joint that was always going to be filled, and it changes how far in the filling goes.

This is also why the head end of every member is left long and cut in place — Chapter 9 again. There is exactly one end of one stick where error is allowed to leave the building as offcut, and the method spends it deliberately.

What the butt cut actually is

One correction, because this project published the wrong version of it and some of that is still out there.

The butt is a **compound** cut. It has a bevel and a mitre.

The bevel is constant at 22.5 degrees — half the sector angle — because it comes from how the log was split rather than from where the member sits in the dome. The mitre takes 3 values across all 120 members.

So the honest claim is not “no compound angles”. It is **three saw settings for a hundred and twenty members**, which is a better claim anyway, and true.

What the pinwheel removes is the shared vertex. It does not remove the mitre.

Nine Processes, Whatever Size You Build

Nine operations turn standing timber into a shell. Not nine categories with sub-steps hiding inside them -- nine setups

The last chapters counted parts. This one counts *verbs*, and the verb list is the shorter of the two.

There are nine things you do to turn standing timber into a shell. Nine. Not nine categories with sub-steps hiding inside them — nine operations, each one a distinct setup with its own tool, its own bench and its own way of going wrong.

You can learn them in an afternoon of reading and be bad at all of them by the end of the first week, which is exactly the right number of things to be bad at.

Here they are, in the order they happen:

fell, rip, crosscut, fold, drill, screw, raise, sheathe, glass

That is the whole shop. Everything in this book that is not thinking is one of those nine.

Why verbs matter more than parts

Parts are things you buy. Processes are things you **get better at**.

A list this short is a list you can actually attack. Nine places to put your attention is a manageable number of places. A list you cannot hold in your head is not a list, it is a fog — and most building methods are a fog, which is why people who are perfectly capable of building things do not build houses.

Fix one process and every dome you ever build gets cheaper. That is not true of fixing one part.

The nine, one at a time

- 1. Fell.** Drop the tree, buck it into sections the length of your longest member plus waste. One chainsaw. The failure is felling into something you did not intend; the fix is the same as it has always been and is not in this book.
- 2. Rip.** Split each section radially into 8 sectors. Froe and mallet, or wedges and a sledge. The failure is a split that wanders off the radius on a section with a knot in it; the fix is to buck around the knot and accept a shorter stick.
- 3. Crosscut.** Cut each stick to length — 70.182 or 62.681 inches — with the compound butt at one end and nothing yet at the other. This is the one cut that can be batched, and Chapter 9 is why. The failure is cutting both ends; the head end is not yours to cut yet.
- 4. Fold.** Assemble three sticks into a panel on the jig, pull the butts up tight, and flush-cut the three heads in place. “Fold” because the triangle closes around itself. Forty times. The failure is doing it in the wrong order and trapping the last stick, which the jig’s stage nine is specifically about.
- 5. Drill.** Pilot and set the threaded inserts in the wedge faces — the ones Chapter 3 argues for. Done on the bench while the panel is flat, which is the only time it is easy.
- 6. Screw.** Join panel to panel, key into seam, hardware into insert. The failure is treating this as structural; it is not. The shell holds its shape because it is a shell.
- 7. Raise.** Ten panels on the base ring, thirty above them, five around the apex. Chapter 10 has the sequence. The failure is bracing too little on the first course, which is the only course that needs it.
- 8. Sheathe.** Close the triangles, then the breather, then the cap. The one watertight layer of Chapter 11.
- 9. Glass.** The openings — and the reason it is last is that an opening is a panel you did not close, so it costs nothing to leave the decision until the shell is up and you can see where the light actually goes.

What is not on the list

No milling. No planing. No jointing, thicknessing, moulding or sanding. No welding, no machining, no hub fabrication, no casting.

Every one of those is a process that some other building method needs and this one does not, and every one of them is a machine, a shed to keep the machine in, and a skill.

The list is short because of the decision in Chapter 1. A frame that is allowed to be lumpy does not need the processes whose only purpose is precision.

Why the list does not grow with the house

Here is the part that matters for anybody thinking about more than one building.

Make the dome twice as wide and the parts list grows: more timber, more sheathing, bigger everything. The *process* list does not change at all. It is still fell, rip, crosscut, fold, drill, screw, raise, sheathe, glass.

Build a second dome and it does not change. Build ten and it does not change.

That is what makes this a manufacturing argument rather than a carpentry one. Every hour you spend getting better at one of nine operations is an hour invested in every dome you will ever build, including the ones you have not thought of yet.

And it is why Chapter 15 can talk about a network of buildings without the argument falling apart: the thing that would normally not scale — the skill — is nine items long.

The honest limit

Two things this does not say.

It does not say the nine are equally hard. Fold is most of the work and rip is most of the danger, and a list that gives them one line each is flattening something real.

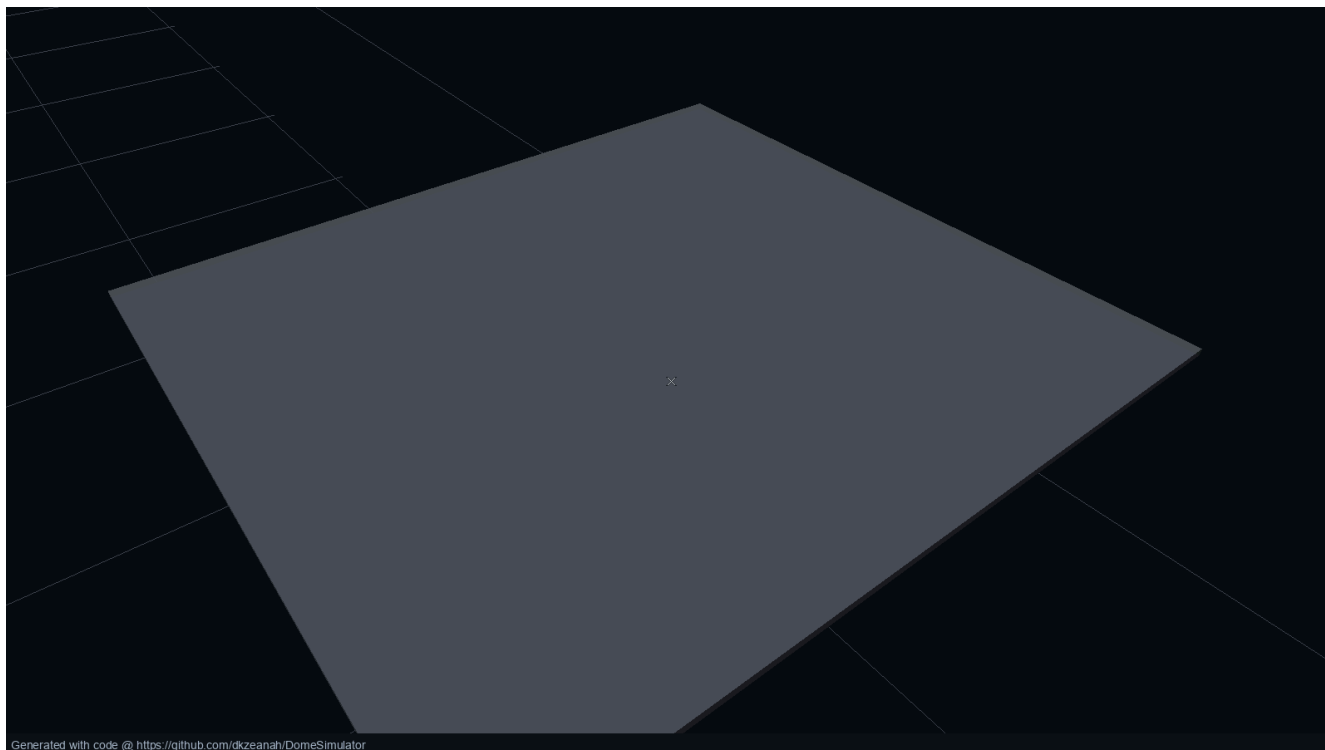
And it does not say nine is a natural number. Count differently and you get seven, or eleven. What is not arbitrary is the *order of magnitude*: this is a method with nine-ish operations, and the alternative is a method with fifty. That difference is the argument, and it survives any reasonable recount.

One Flat Board, Forty Identical Panels

The only thing in the whole build that has to be true is the bench, and everything else is measured from it

The only thing in this entire build that has to be true is a sheet of plywood on two sawhorses.

Everything else is measured from it, and nothing on it is adjusted between panels — so panel forty is panel one.



Generated with code @ <https://github.com/dkzeanah/DomeSimulator>

One flat sheet on sawhorses, and nothing else.

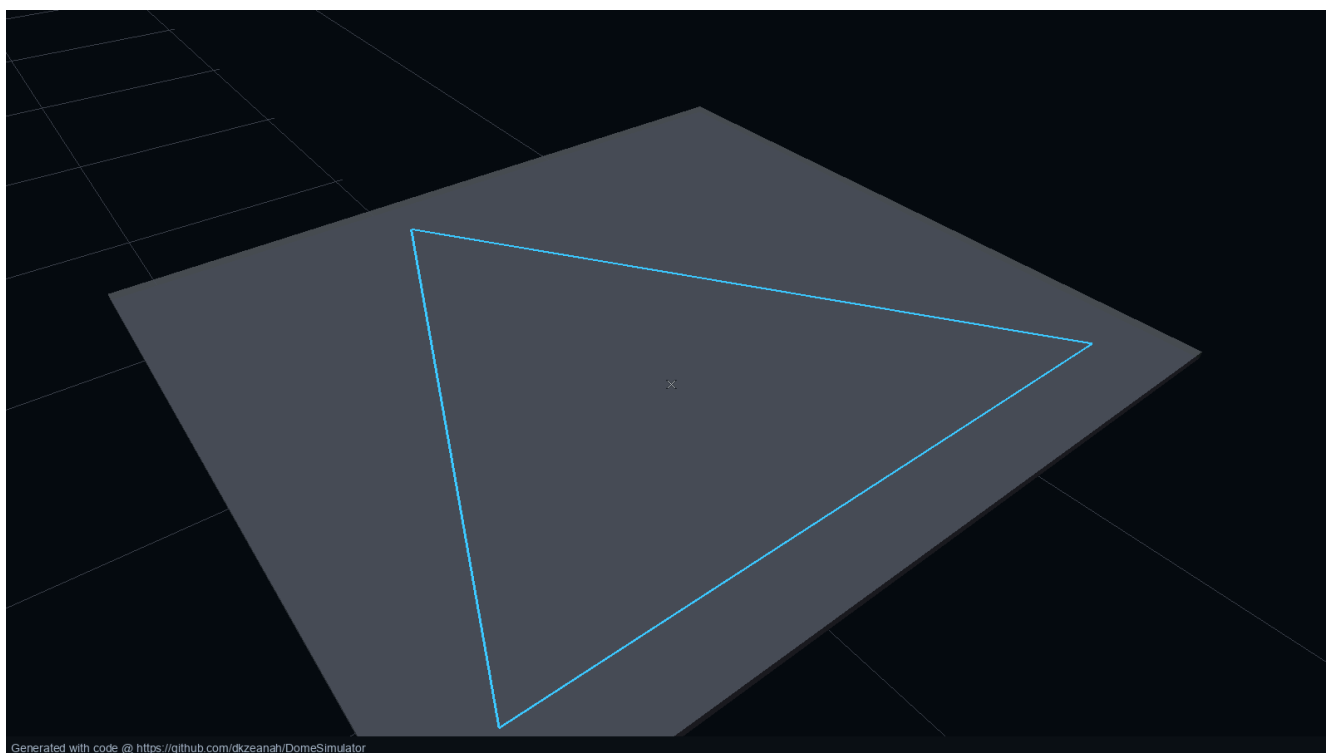
Building the jig

1. The bench

One sheet, flat, level enough that a marble does not run. Flatness is the whole specification: every angle below is measured from this surface, so the panel cannot wind or twist as long as the board does not.

Check it with a straightedge on both diagonals. If it rocks, shim the horses, not the sheet.

2. The triangle, struck once



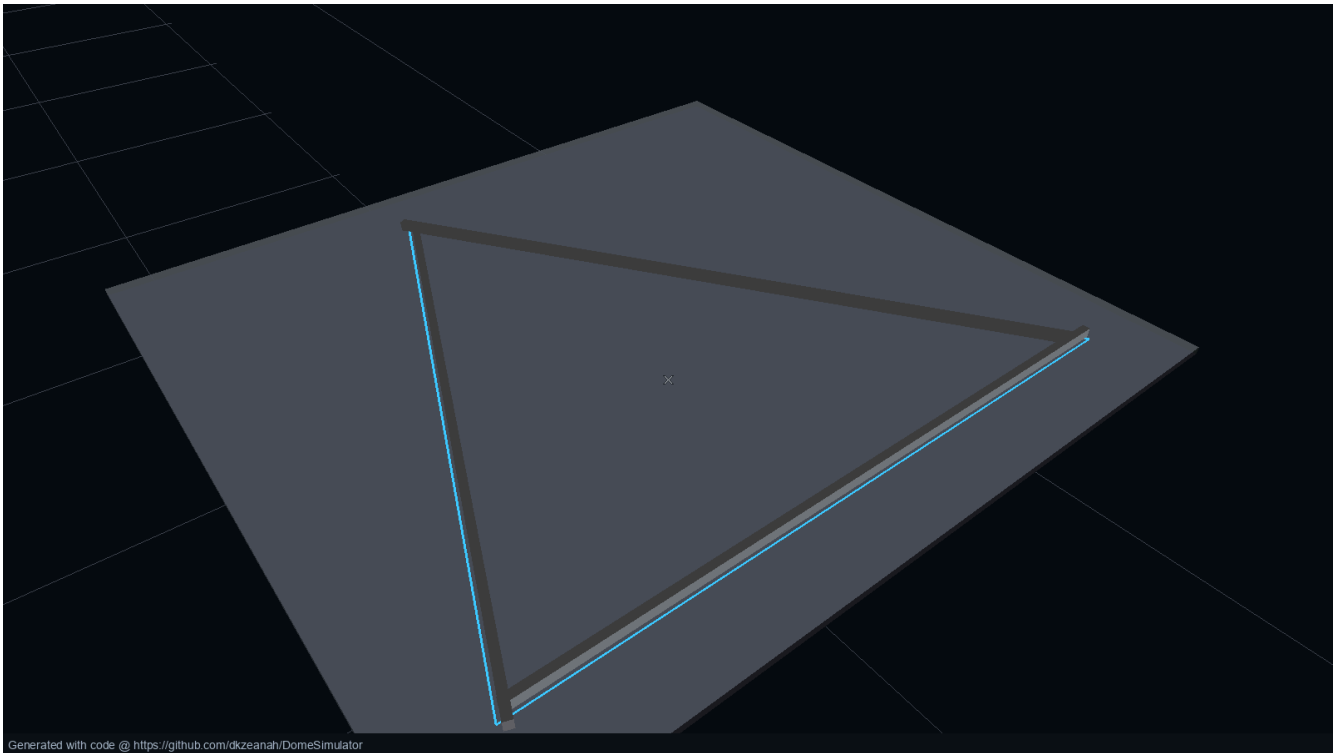
The panel's triangle, struck once from the solution and never measured again.

Snap the panel's mathematical triangle onto the board — the 72.000 and 63.670 inch sides of Chapter 6.

Strike it **one time**, from the geodesic solution, and never measure it again. Forty panels laid out from three lines struck once is a different proposition from forty panels laid out from forty tape measurements, and the difference is the entire reason this works.

The tool prints this triangle full size as a drawing you can lay on the sheet. Chapter 18 says how.

3. Axis rails



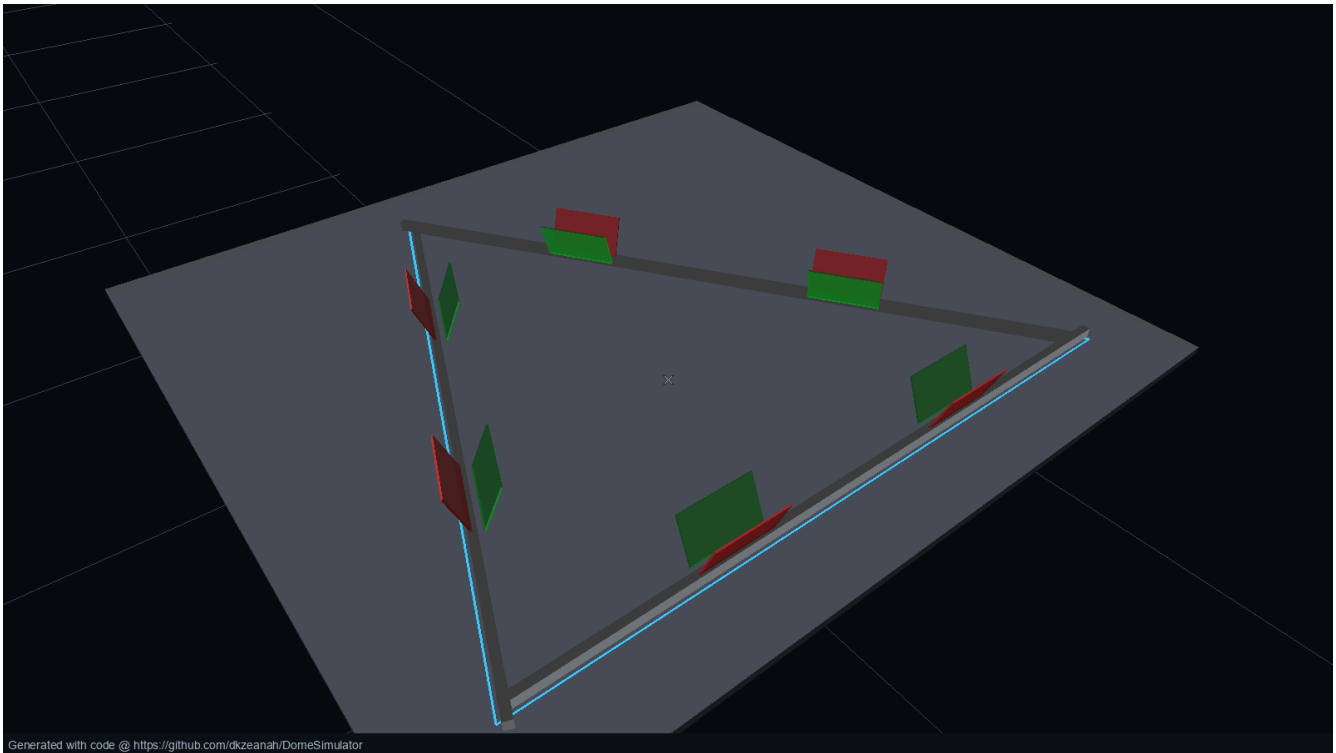
Three cleats, one per member, each parallel to that member's point-axis.

Three low cleats, one per member, each set parallel to that member's point axis and just to the seam side of it.

These fix **position across the panel**. Push the stick against its rail and the sector's point line sits exactly where the solution says.

The rail references the sawn point, never the bark. A lumpy log does not move the joint.

4. Red and green fences



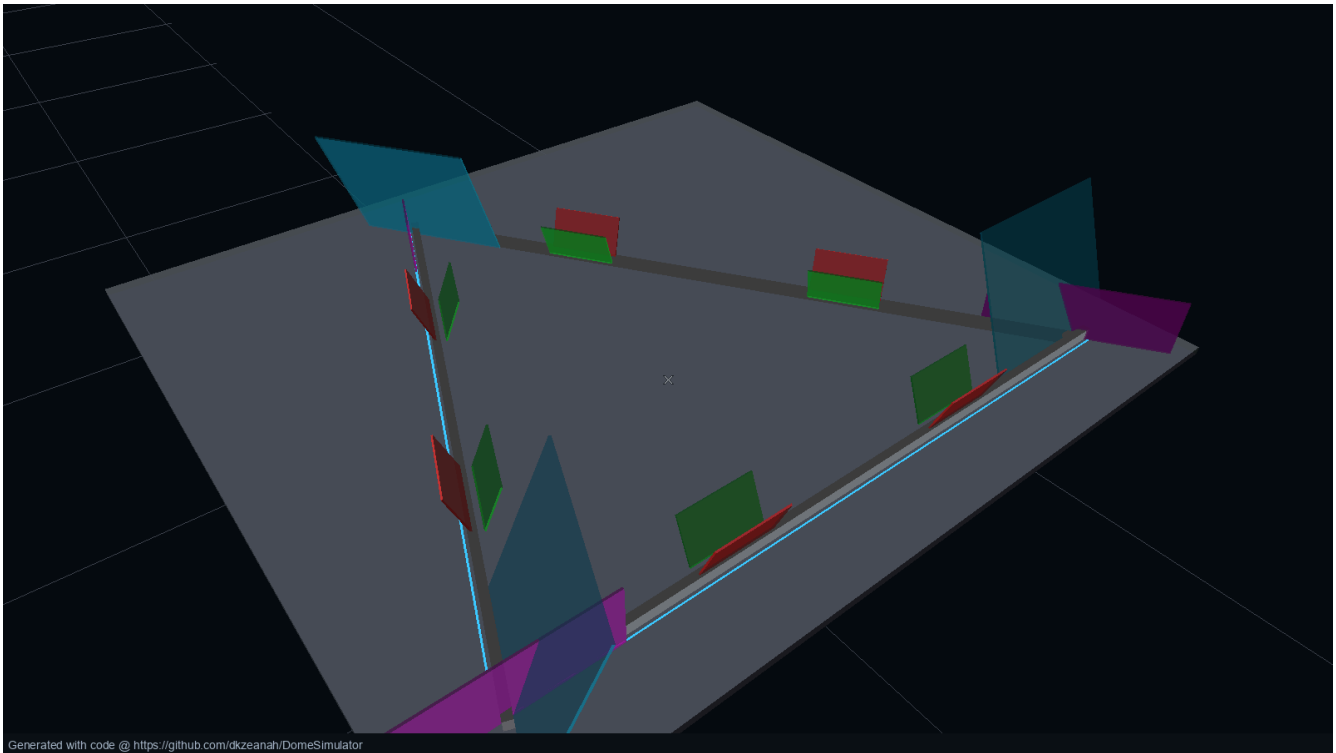
Red to red, green to green, or the stick does not seat.

Two short cradle plates per member, standing against the two sawn radial faces of the wedge.

These fix **rotation about the stick's own axis**, and they are the reason you cannot get a member in the wrong way round. A sector dropped in rotated ninety or a hundred and eighty degrees still *looks* fine. These plates make it not fit.

Mark them. Red side to red plate, green side to green plate. It feels childish for about two panels and then it saves you one.

5. The two cut planes



The magenta plate is the flush-cut fence; the cyan is the face the next butt lands on.

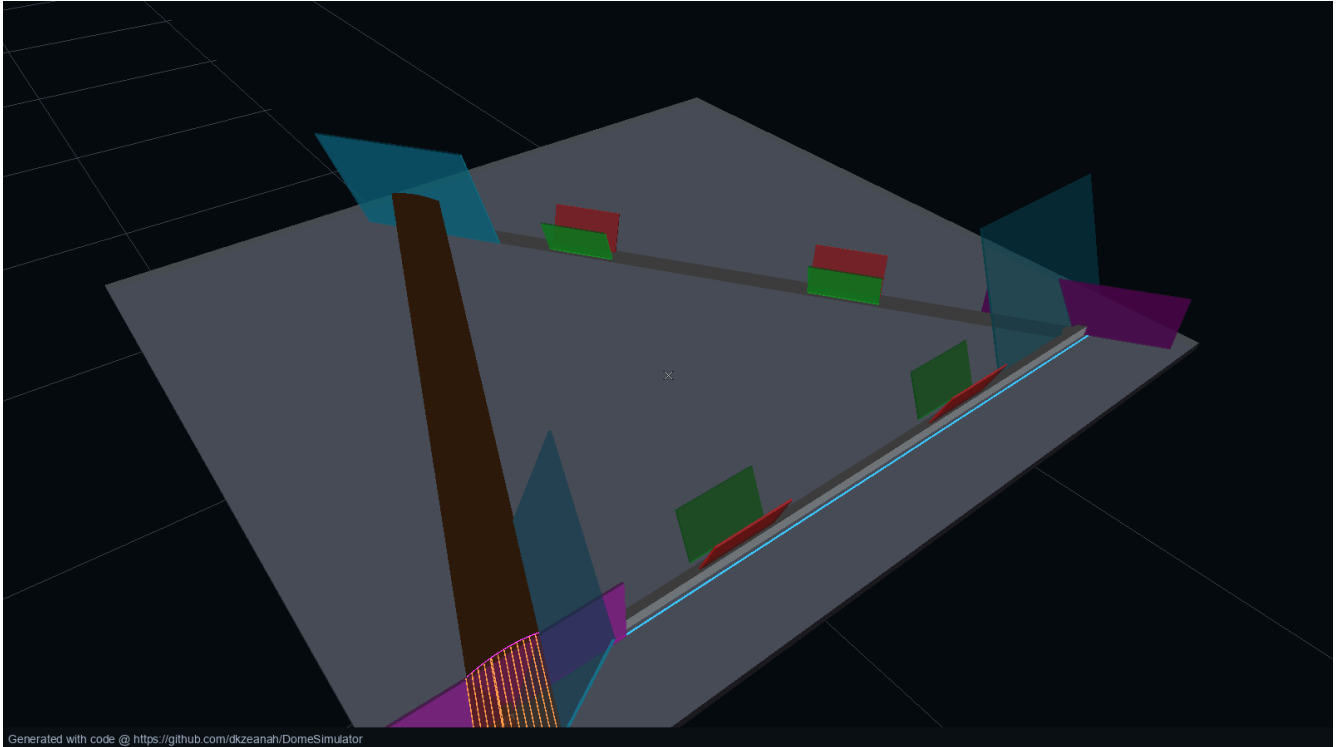
At each head end, a plate standing in the plane of the triangle's edge. That is the flush-cut fence: a saw run flat against it cuts the head to exactly the right plane without anybody setting a bevel or a mitre.

At each butt end, a plate in the plane of the receiving face the next stick's butt will land on.

That is the fixture. A sheet, three cleats, six cradle plates, six guide plates. Nothing on it moves again.

Making one panel

6. The butt cut, off the jig



The one cut made before assembly, and the only one that can be batched.

One stick, still full length at both ends, laid up to the cyan plate. Cut the butt.

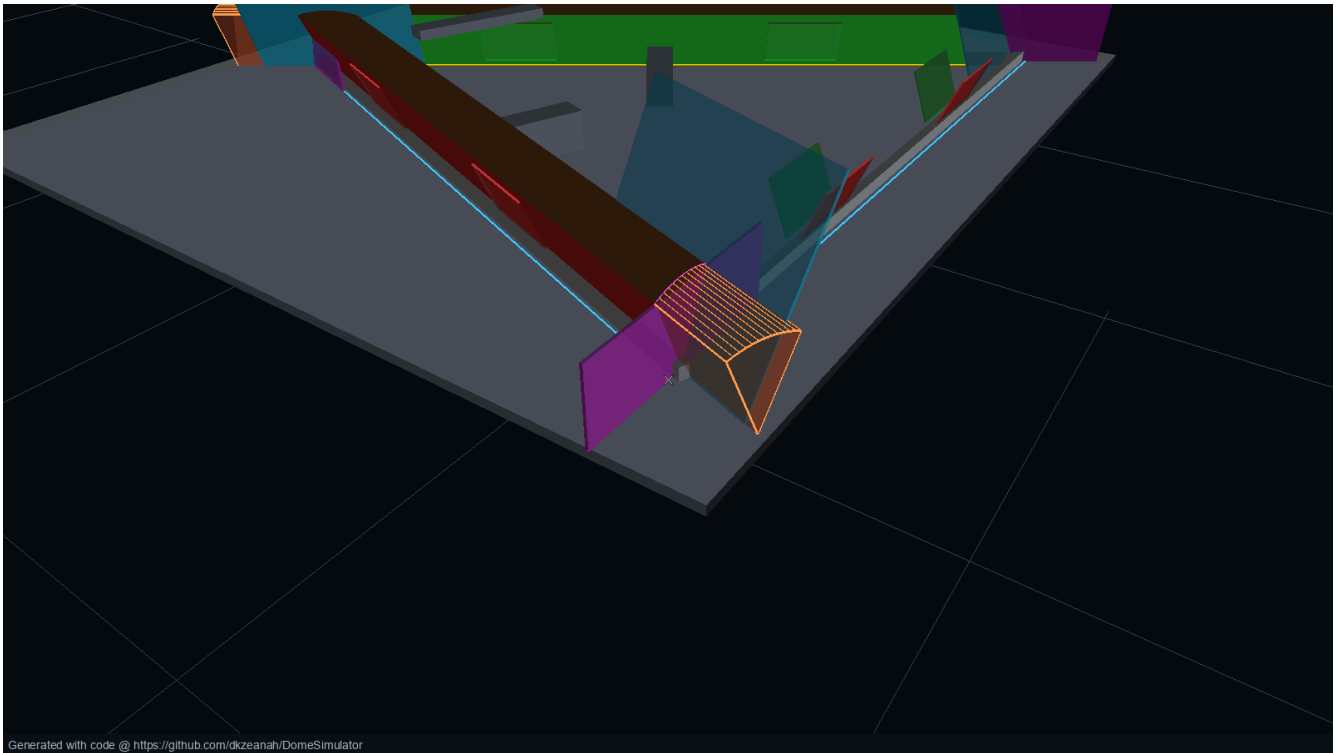
This is the **only** cut made before assembly, and every stick of the same family takes the same one — so it runs through a saw bench in a batch rather than being fitted one at a time. Sixty A butts, then sixty B butts.

7. First stick down

Butt now cut, dropped against its rail and its two fences. The head end is still long and hangs off the board.

Two ends, two completely different operations. The butt arrives finished. The head arrives long on purpose and you do not touch it yet.

8. Second stick, butt onto side

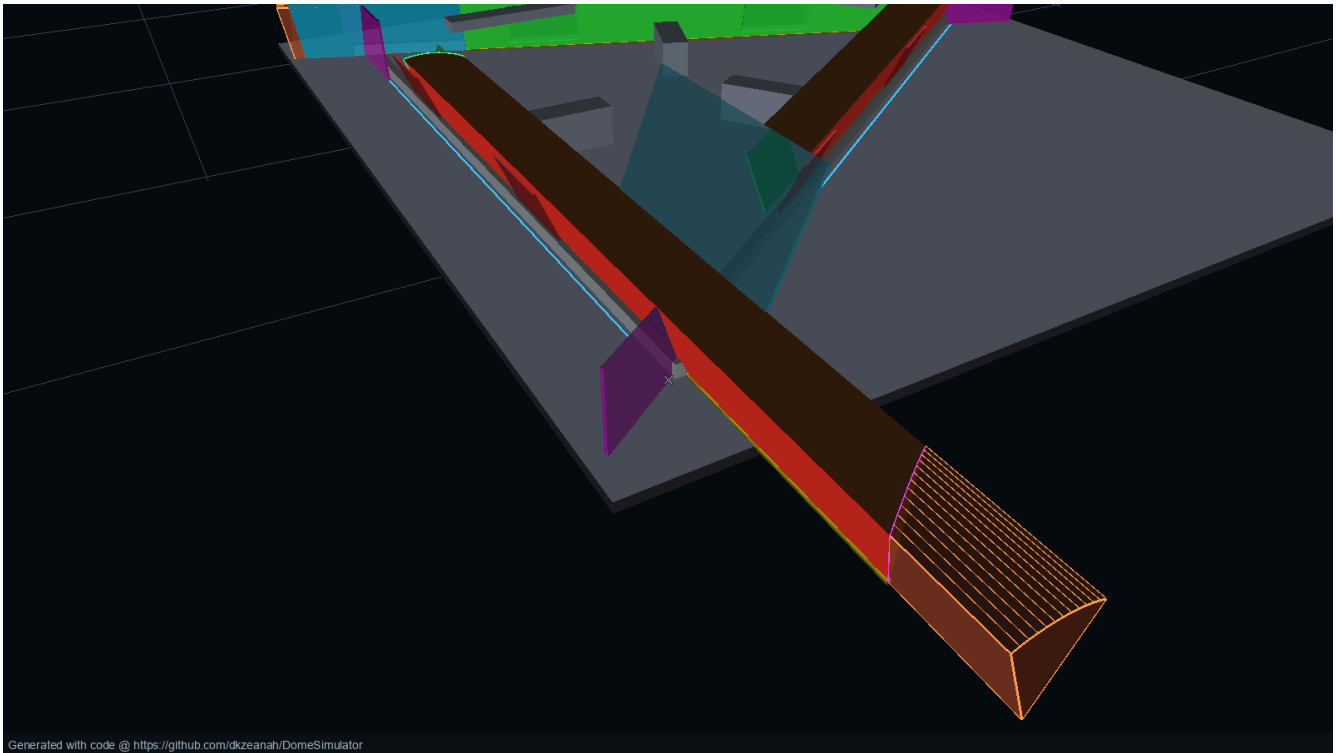


End to side: a flat sawn face on a flat sawn face.

The second stick lands so the first stick's butt bears on its **radial face** — on the side of it, not on its end.

Nothing is mitred to meet at a point. A flat sawn face is pressed onto a flat sawn face, which is what a split wedge already has without any further machining.

9. The last stick slides in



The last stick is captured at both ends, which is why the head is left long.

The third stick is captured at **both** ends once the other two are down, so it cannot drop in from above. It comes in along its own axis, sliding home into the pocket.

This is the one awkward move in the whole method, and it is why the head end is left long: the extra length is what gives the last stick somewhere to come from.

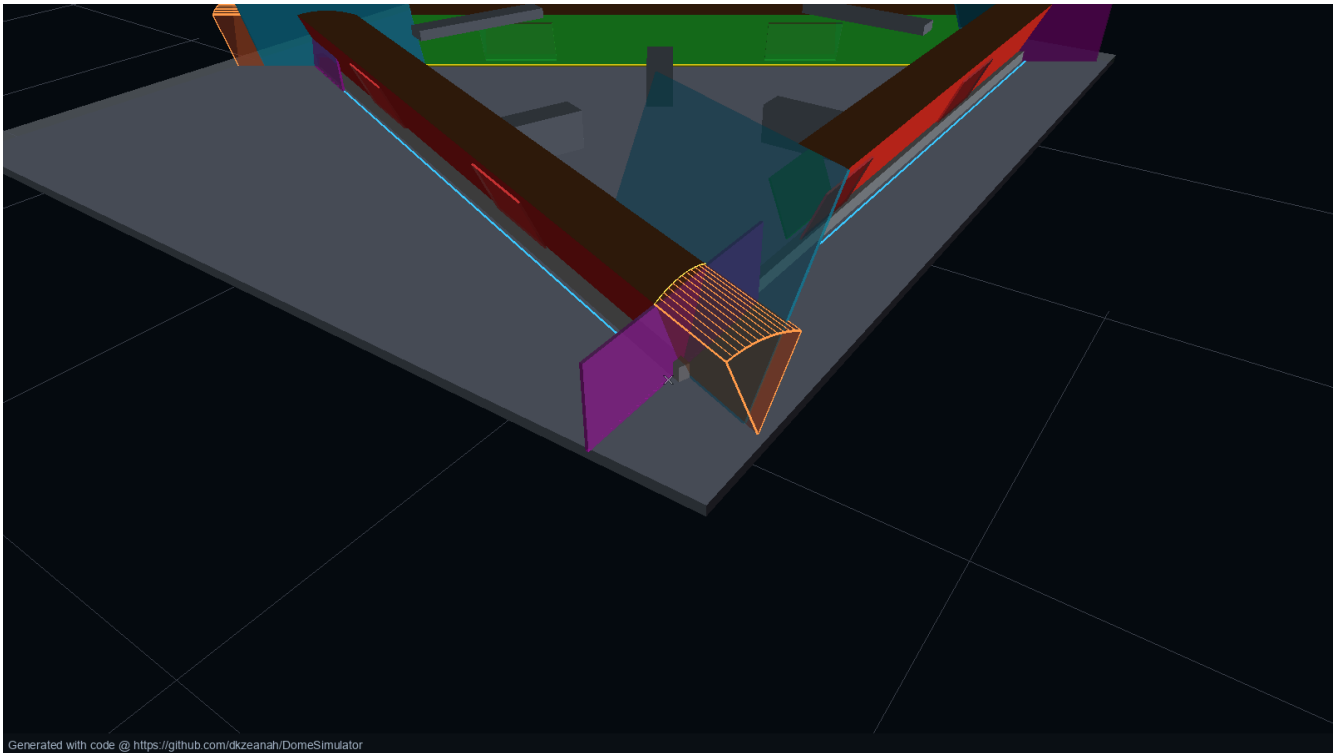
If you have trapped it, you put the sticks down in the wrong order. Take one out and start again; you will do this once.

10. Pull it up tight

Three butts bearing on three sides. The loop is closed, the frame holds its own shape, and the jig is now only keeping it flat while you finish.

All three heads still hang off long.

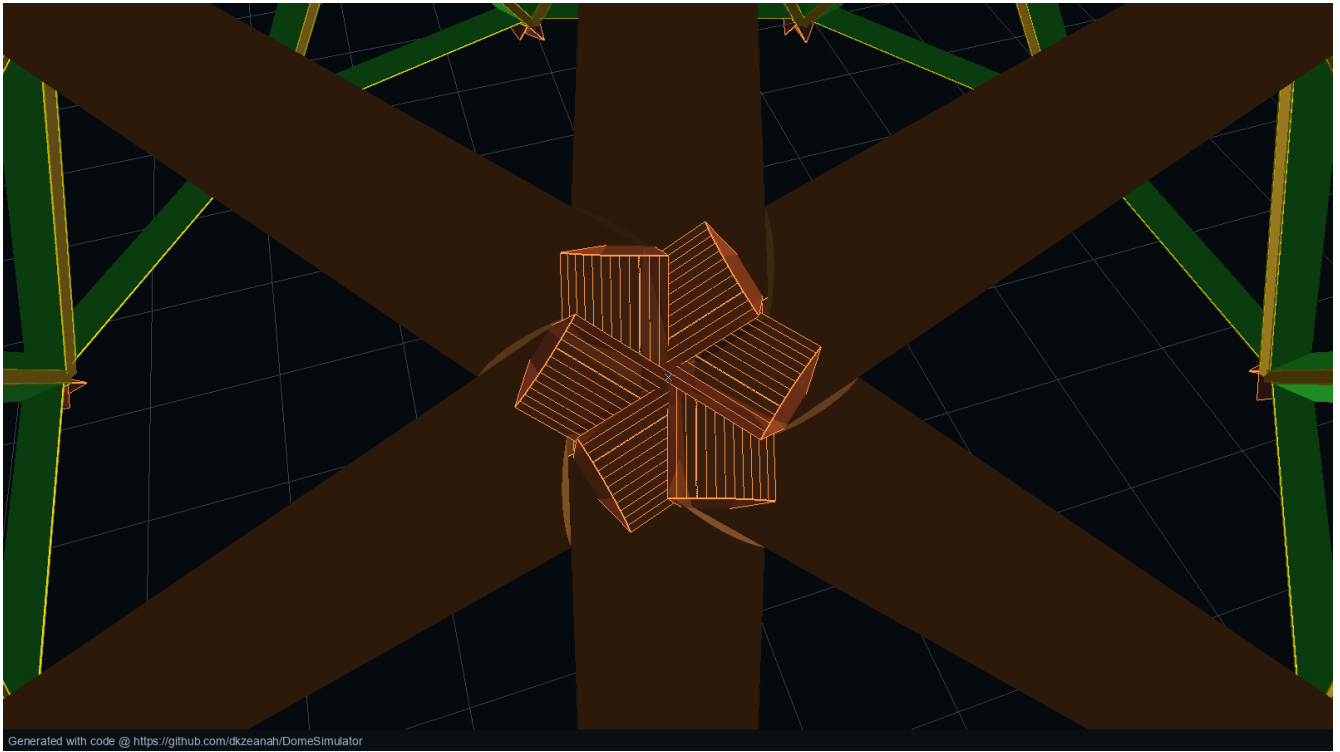
11. Flush-cut the three heads



The head is never measured. It is cut in place.

Saw each head back to its magenta fence. The stock hanging past the board falls off.

This is the whole trick. The head is never measured. It is cut in place, to the plane the neighbouring member has already established — so error in stock length, or in the butt angle, or in how round the log was, leaves as offcut instead of accumulating around the triangle.



Uncut, every head trespasses across the vertex.



And after.

Those two pictures are the same corner of the same dome. In the first, every head still carries its stock and they trespass straight across the geodesic vertex into the neighbouring panel. In the second they end inside their own triangle and nothing crosses.

12. Panel off the jig

A finished triangular frame, its three corners clean inside the mathematical triangle. The jig is empty and unchanged.

Drill the inserts now, while it is flat — Chapter 3 — and sheathe it now if you are sheathing on the bench, which you should be.

Then do it 40 times, in two families: 10 equilateral and 30 isosceles.

What the jig is really buying

Repeatability, obviously. But more than that: it moves the hard work indoors and off the ladder.

Every operation above happens at waist height, on a bench, with both feet on the ground, in whatever weather. The part of dome building that everybody pictures — the awkward bit up a ladder holding something heavy at an angle — happens once, in Chapter 10, and takes three days out of a fortnight.

PART 4

The Skin

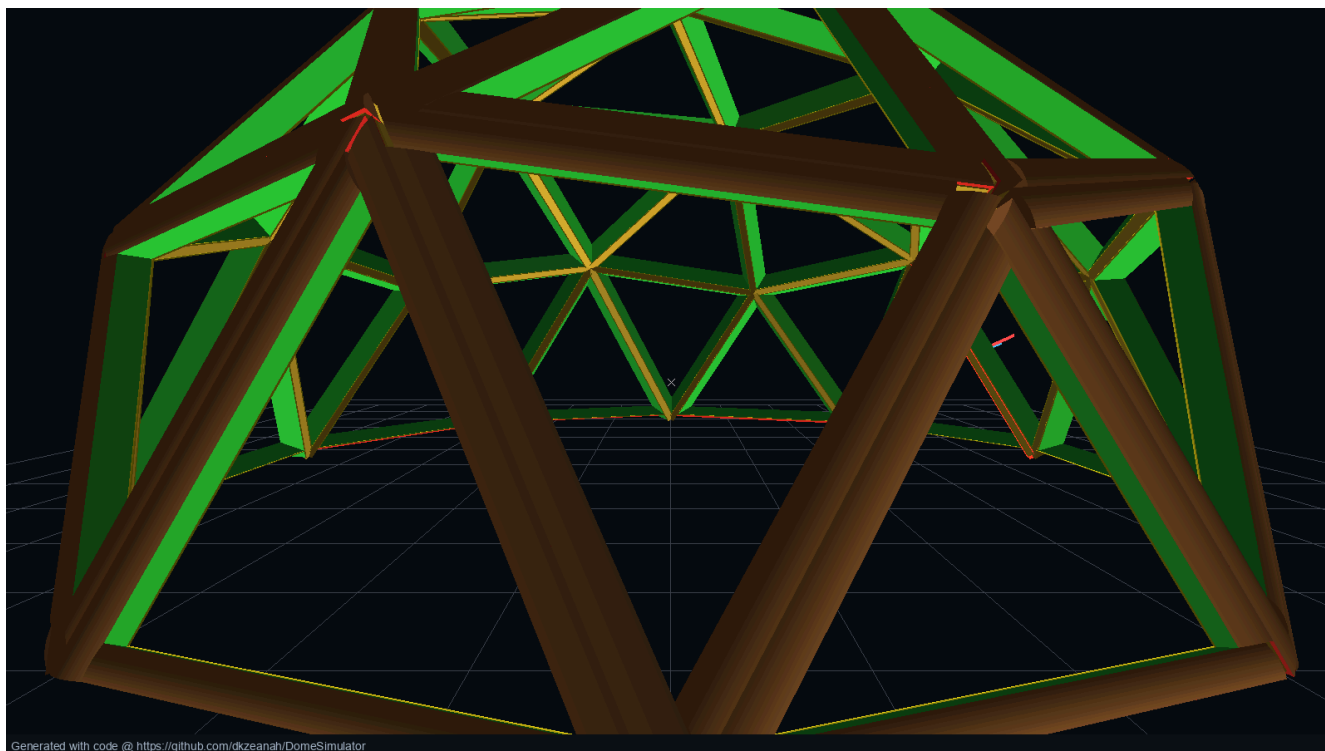
“A frame that is out by an inch stands for fifty years. A skin that leaks for one winter takes the frame with it.”

Raising the shell, the single watertight layer over it, and the hats that go on afterwards -- the part of the building this book says matters most.

Raising It

Ten panels on the ring, thirty above them, and five around the apex -- in that order, and braced until the second course closes

Forty panels are stacked against a fence. This part takes three days and it is the only part that looks like building a dome.



The ten panels that stand on the ring.

Before anything goes up

Set the base ring out first. A regular decagon, 10 sides, each one an A chord — so each side is 72.000 inches and the perimeter is sixty feet. Chapter 13 is what it sits on.

Get this right and the rest is assembly. Get it wrong and every course above it inherits the error, because the base is the one place in this building where error does accumulate.

Check it by diagonal, not by angle. Opposite corners of a regular decagon are a known distance apart; measure two of them and compare.

Sort the panels into two piles. 10 equilateral, 30 isosceles. Mixing them up costs an hour on the second course.

Have more bracing than you think. You will need it for about six hours total and you will need all of it at once.

The first course

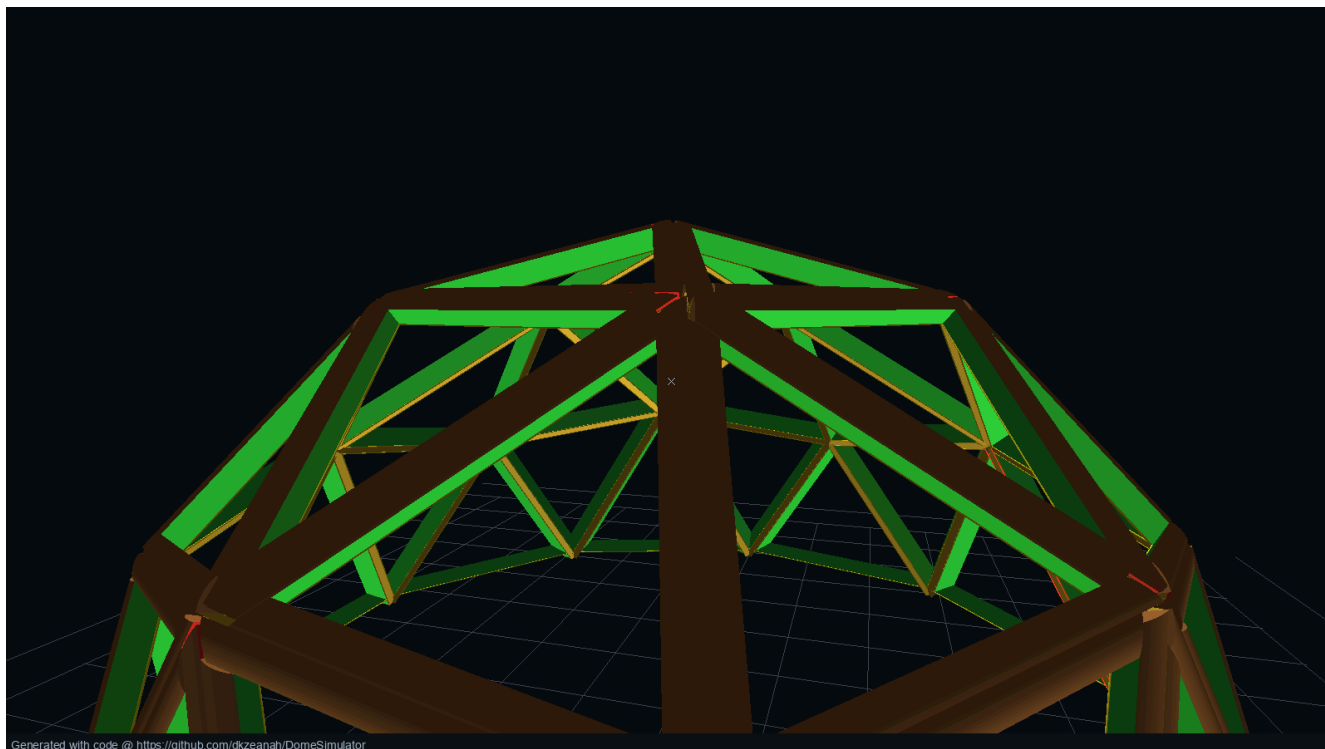
Ten panels stand on the ring. They lean inward from the moment they go up, which is the first surprise: a dome is not a wall that gradually curves, it is leaning in from the ground.

Stand two adjacent panels, key the seam between them, and brace the pair externally. Then work round. Each new panel keys to the one before, and the run of them wants to fall inward until the ring closes.

The ring closing is the event. Until the tenth panel keys to the first, you have a curved wall held up by props. After it, you have a structure.

Do not take the bracing out yet.

The second course



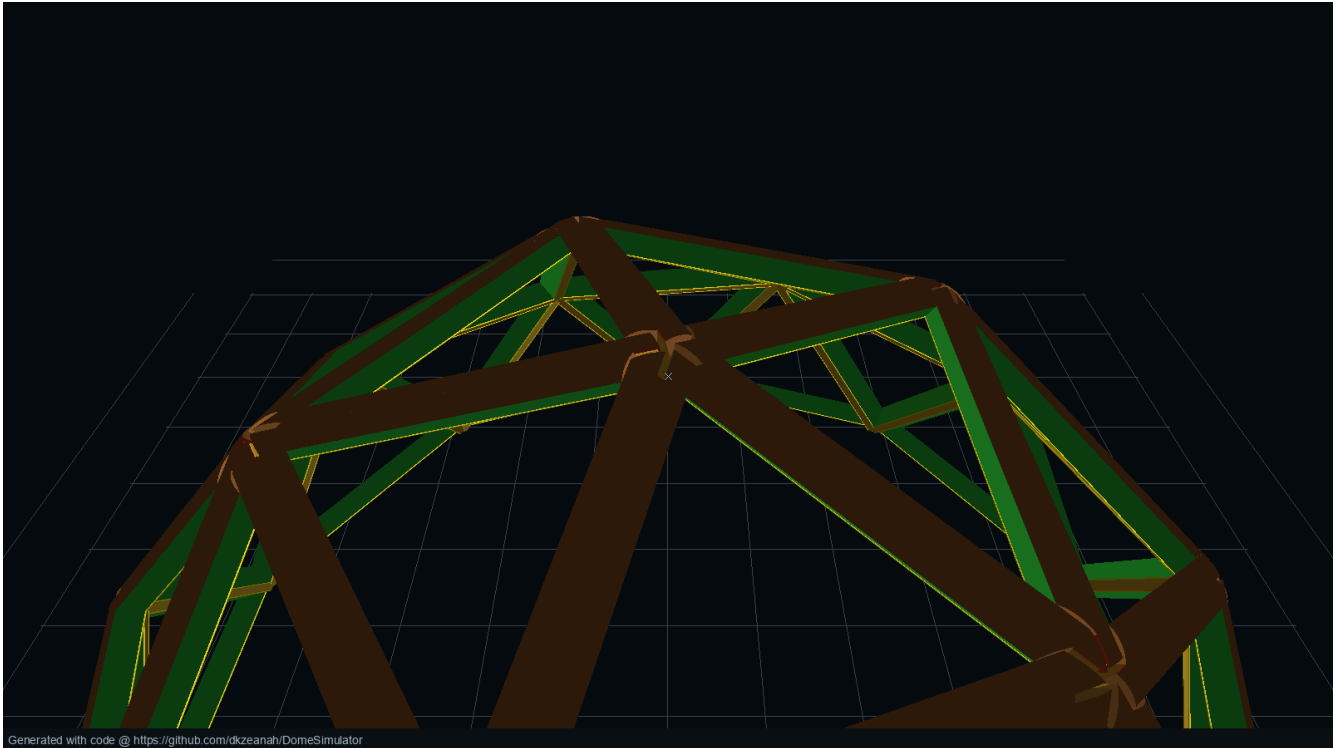
The band above: where the dome starts holding its own shape.

Thirty panels, leaning further in, each one keying to two below it and to its neighbours either side.

This is where the dome starts holding its own shape — and it is where the bracing can come out, once the course is closed all the way round. Not before. A half-closed second course is the least stable the building will ever be, because you have added weight and height without adding the ring that resolves it.

Work from opposite sides toward a closure point rather than round in one direction. It keeps the load symmetric and it means the last panel of the course is a panel rather than a wedge-shaped gap you have to persuade.

The apex



Closing the top.

Five panels around one vertex.

This is the last thing to go in and the only part of the assembly that has to be *held* rather than propped. Two people, or one person and a strap from below.

If you have built the mast of Chapter 17, the apex ring is already up there and the last five panels close around it. If not, the apex is five keys and a cap.

What goes wrong

The last panel of a course does not fit. Almost always the ring is out, not the panel. Check the base diagonals before you start cutting anything.

A seam gaps at the outside. The key is not driven far enough. It is a wedge; drive it.

A panel will not key to its neighbour. One of them is in upside down. This happens with the isosceles ones, whose two short sides look alike. Mark the long edge of every BAB panel on the jig, in

the shop, before it ever comes outside.

Then stop

The frame is up. It is weather-tight in the sense that a bird cage is weather-tight, and it will stand like this indefinitely.

Do not stop here for a winter. That is the failure Chapter 3 describes, and this is the moment it happens: the frame is standing, it looks finished, it is intoxicating, and the skin is the expensive part.

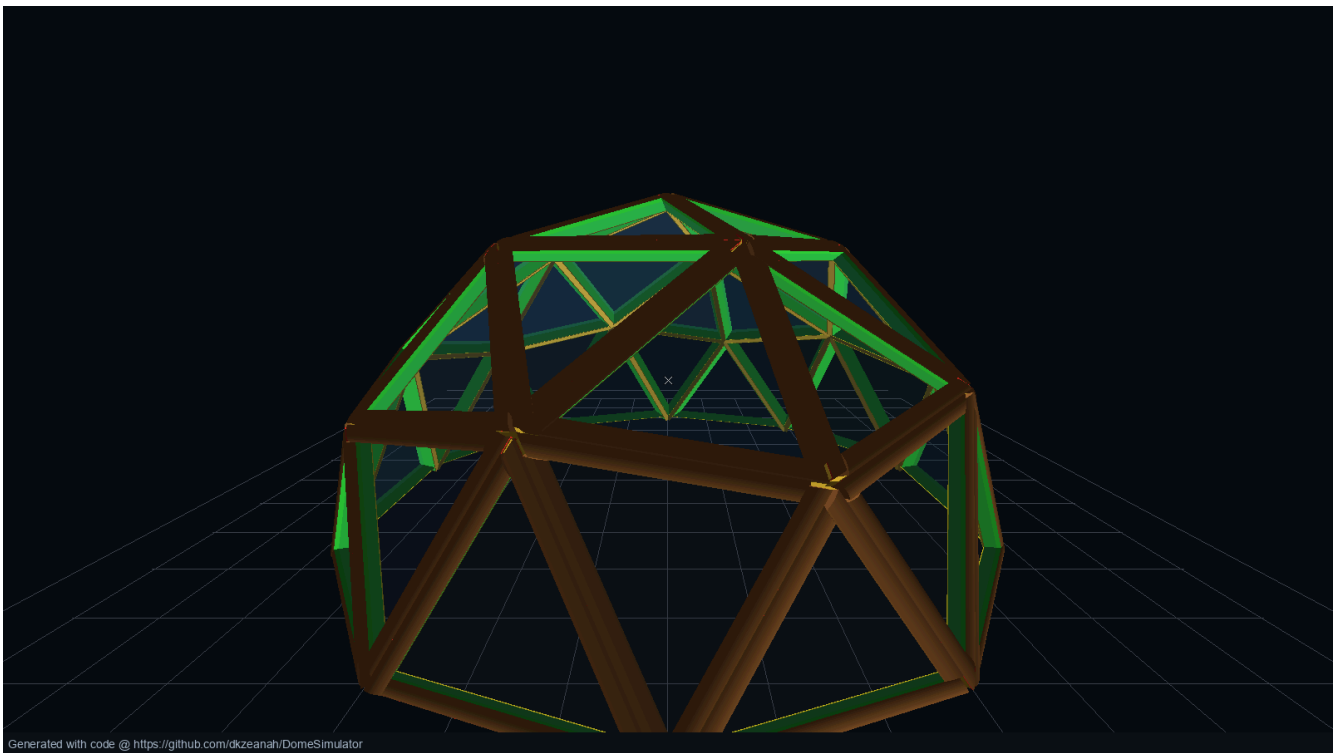
The skin is the part that matters. It is the next chapter, and it is the reason all of this stands up in thirty years' time instead of in three.

One Watertight Layer, and Only One

Two impermeable skins with insulation between them is a trap, and this project shipped that design before it noticed

Put a waterproof layer on the outside of a building and you have kept the rain out.

Put a second one on the inside of the same wall and you have built a bag.



The frame with its skin on.

What this book used to say

The shell in an earlier version of this design had a waterproof membrane *under* the panels and a rain-slick cap *over* them, with the insulation between the two.

That is a vapour trap. Water gets into every wall — from the weather, from the air inside, from somebody's breath — and the only question a wall design answers is where it goes next. A wall with an impermeable layer on both sides has no answer. Moisture arrives, cannot leave in either direction, and condenses on whichever surface is colder that night.

The old design listed this as an unresolved concern, in its own documentation, for months. Writing it down is not the same as fixing it.

The fix is the subject of this chapter and it came from a one-line note: there should only be one watertight layer, on the outside, like a shower cap.

The stack, outside in

The cap. One continuous rain-slick layer over the entire shell. Welded seams, no fasteners through it, strapped down to ground anchors at the base vertices. Its only job is liquid water and it is very good at that job because it is doing nothing else.

A vented gap. Three quarters of an inch between the cap and what is under it. This is the part people skip and it is the part that makes the rest work: any moisture that gets past the cap, or arrives from inside, has somewhere to go and something moving through it to take it away.

The breather. A vapour-permeable sheet at around ten US perms. Not a poly sheet. It stops bulk water and lets vapour through, which is the exact opposite of what a waterproof membrane does and is why the distinction matters more than the price difference.

The quilted layers. As many as you have. Chapter 12.

The panels. Wood. Structural, not weatherproof, and not asked to be.

The frame. Dry, because everything above it is doing its job.

Which way the vapour goes

Warm air holds more water than cold air. Inside your building the air is warm and full of you, your cooking and your breathing. Outside it is cold.

So vapour moves outward through the wall, and it will condense at whatever point in the wall is below its dew point. The design question is: **can it get out before it gets cold?**

With a breather and a vented gap, yes — vapour passes through the permeable layer into moving air and leaves. With a second impermeable layer, no. It stops at the membrane, at a temperature partway through the insulation, and condenses there, inside the insulation, where you will not see it for years.

That is the whole argument, and it is why the cap is the only waterproof thing in the building.

What it costs

A bare cap — no quilted layers at all — is \$3,265 on this dome. With layers it climbs slowly; the shell in the standard article is \$3,265.

The alternative is a laminated hull: glass fibre in resin over the whole envelope. Genuinely waterproof, sixty years, and \$21,513 for the same dome against \$12,036 — a real premium for a real difference.

Chapter 16 prices four laminates and the cap side by side. The short version: the hull is better at being waterproof and worse at everything else, because it is permanent. It cannot be lifted off, it cannot be upgraded, and when it fails it fails as a repair job on a bonded shell rather than as a component you unstrap.

The three things a cap has to get right

It has to be one piece, or welded. A cap made of overlapping panels is just a roof with more seams, which is the problem it exists to remove.

It has to be strapped, not fastened. Every fastener through a waterproof layer is a hole in a waterproof layer. Straps over the outside to anchors at the base put the load into the ground without putting anything through the membrane.

It has to be bigger than it needs to be. Chapter 3: a cap cut to fit a bare shell will not go over a shell wearing two quilted layers.

The part that is not solved

Ultraviolet.

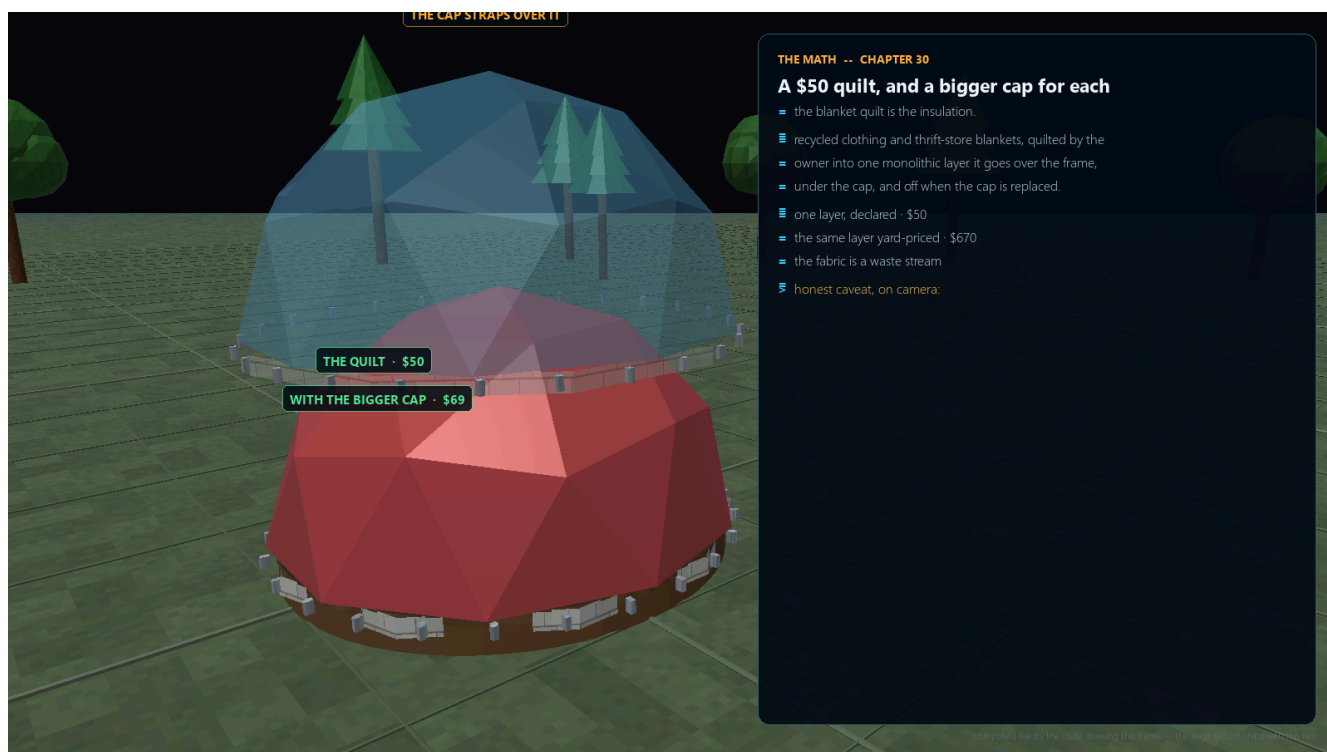
A rain-slick outer layer sits in the sun for its whole life, and that is what eventually finishes it — not water. The cap is a consumable on a decade-ish timescale, and the design's answer is that it is a consumable **by design**: unstrap it, pull it off, put the next one on, in a day, with the building occupied throughout.

That is a genuinely different answer from a roof, and it is the right one for a component whose failure mode is known and gradual. It is still a cost, it is still a thing you will do four or five times in the life of the frame, and this book would rather say so than let you find out in year twelve.

A Dome Is a Head, and It Wears Hats

Bare skin, then layers, then one rain-slick shell over the lot -- the way you dress for the Arctic, and for the same reasons

I learned how to dress a building standing on the outside of a ship in the Arctic Circle, on forward lookout, at night.



A quilt, and a bigger cap for each one.

What you wear out there

Nobody wears one warm coat. You wear layers — several of them, each thin, each doing one job — and on top of all of it goes a pumpkin suit: an immersion suit, bright orange, entirely waterproof, and utterly useless as insulation on its own.

The suit does not keep you warm. The layers keep you warm. The suit keeps the layers *dry*, and a dry layer is worth several wet ones.

The other thing about that arrangement is that it is adjustable in the middle of a watch. Too cold, add a layer. Too warm, lose one. You never take off the suit.

That is the whole shell design in this book, and I did not arrive at it by reading about vapour barriers.

The dome as a head

A dome is a head. It is round, it is the top of something, and it wears hats.

It can wear a bare skin — nothing at all, a frame with panels on it.

It can wear a toboggan: a quilted layer, then another, stacked, each one adding warmth.

And over all of it, always, one shower cap. Waterproof, thin, useless as insulation, and the reason every layer under it is dry.

People understand this instantly in a way they do not understand “vented rain screen with a vapour-permeable weather-resistive barrier”, which is the same idea in a language that makes people stop listening.

Each hat is a size up

This is the part that is arithmetic rather than analogy.

A layer goes *over* the layers already there, so it is bigger than the one under it. Not a lot bigger — the shell is only a few inches thicker each time — but the areas compound, and by the seventh layer the outermost is meaningfully larger than the innermost.

So “one quilt costs fifty dollars” is a sentence about the first quilt. The model works out each layer’s own area, and the cap that goes over all of them is sized for the stack you intend to end up with rather than the stack you have.



The envelope gaining a layer a winter.

It gets warmer every winter

Here is what the layering model buys that a conventional wall does not.

A conventional wall is insulated once, when it is built, to whatever you could afford that year. Changing it later means opening the wall.

This one is insulated from the outside, in increments, forever. Each autumn you can add a layer — an evening's work with a sewing machine and a box of old clothes — and the building is warmer than it was last winter. The cap goes back over the top and nothing else about the building changes.

A house that improves as you live in it is a different relationship from a house that is as good as it will ever be on the day you move in.

And the quilts are made of clothes

The insulating layers in this design are quilted from recycled textiles — clothing that would otherwise be waste.

That is partly an economic argument and it is mostly a participation one. It is the only part of this entire method that somebody can do at a kitchen table with no land, no saw and no site. A layer is a real object with a number on it that goes on a named building, and it takes an evening.

The arithmetic is in the model: how many garments, how many pounds, what R-value the stack reaches, and what a layer is worth to the person who sews it for somebody else's dome. That last one has to be a real number or this is not a network, it is a request for free labour.

Where the analogy stops

Two places, and both matter.

A person moves and a building does not. Your layers work partly because you are generating heat inside them and moving air through them. A building generates less heat per unit of surface than you do and does not move at all, so it relies on the vented gap of Chapter 11 to do what your movement does.

You take your suit off indoors. The dome does not. The cap is on it in July, in the sun, which is the ultraviolet problem at the end of the last chapter and the reason the cap is a consumable.

Neither of those breaks the model. They are the two places where a builder who reasons from the analogy alone will get it wrong, so they are written down.

PART 5

The Floor and the Ground

“Everything you own sits on the floor, and the ground is the one part of the building you cannot take with you.”

The second thing that has to be right, the five ways to build it, and the argument for standing on somebody else's ground.

The Floor Is the Other Thing to Get Right

It carries everything you own, you touch it all day, and it is where the ground sends its water

You will spend more hours in contact with the floor than with any other part of this building, and it is the only part in permanent contact with the ground.

Both of those are reasons to spend money here rather than on the frame.

What “right” means

Dry. The ground is wet. It is wet under a slab, under a deck, and under gravel, and it will move moisture upward into anything that touches it unless something stops it. A capillary break — gravel, a membrane, an air gap — is not optional and it is cheap at the time and impossible afterwards.

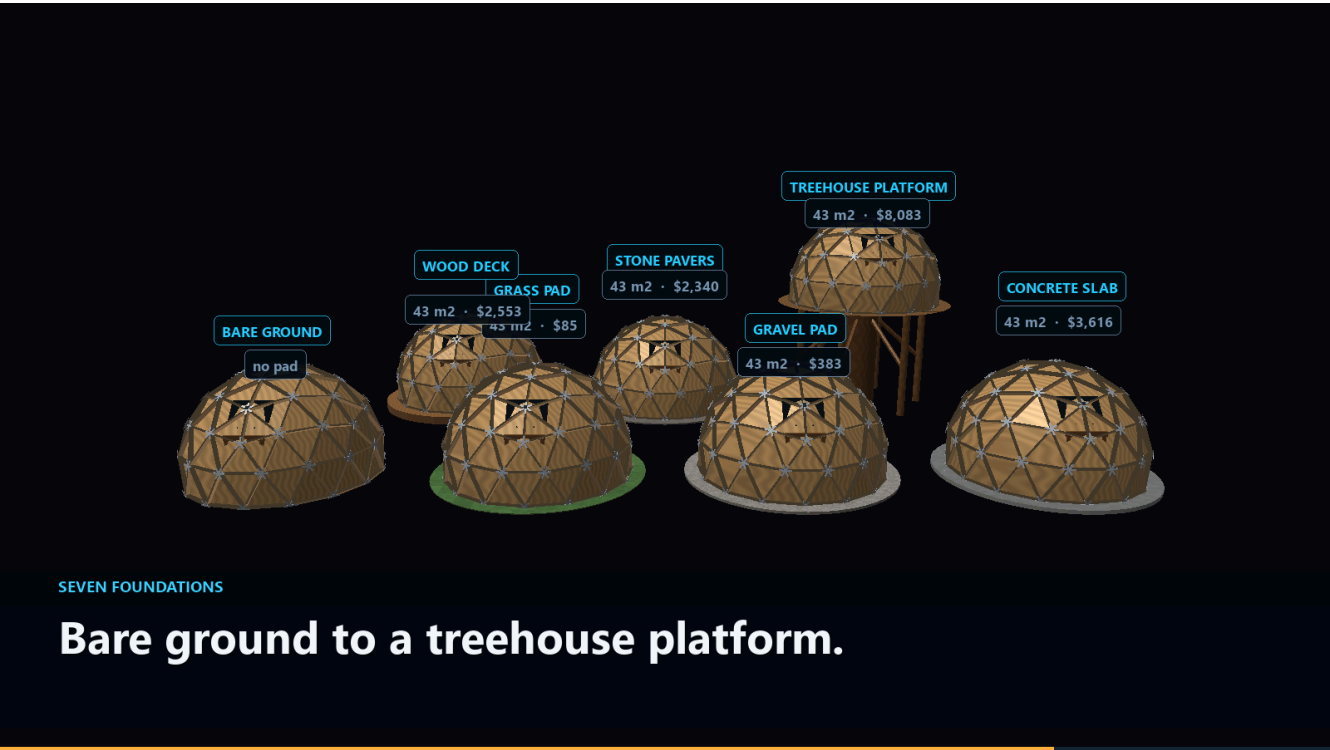
Warm underfoot. A cold floor makes a warm room feel cold, because you are touching it and you are not touching the air. This is worth more comfort per dollar than the same money spent on the walls.

Flat and stiff. You notice a floor that flexes every single day.

Level, to a few millimetres. Not because the dome needs it — the dome would stand on a slope — but because everything you put in the dome needs it.

Five ways to do it

Priced on this footprint, taken off board by board rather than rated per square foot:



Bare ground to a treehouse platform.

compacted gravel base	\$443
full concrete slab	\$2,468
framed deck on piers, sealed	\$2,916
concrete ring with a timber middle	-- see the table
framed deck, ply and epoxy over it	-- see the table

The staple is the framed deck on piers at \$2,916: it is dry by construction, it is warm because there is air under it, it is repairable, and it does not require a concrete truck to reach your site.

The gravel base at \$443 is not a floor. It is a prepared ground that a floor can go on. It is in the list because it is what a lot of people actually start with, and it is honest to say that it is a starting point rather than a cheap answer.

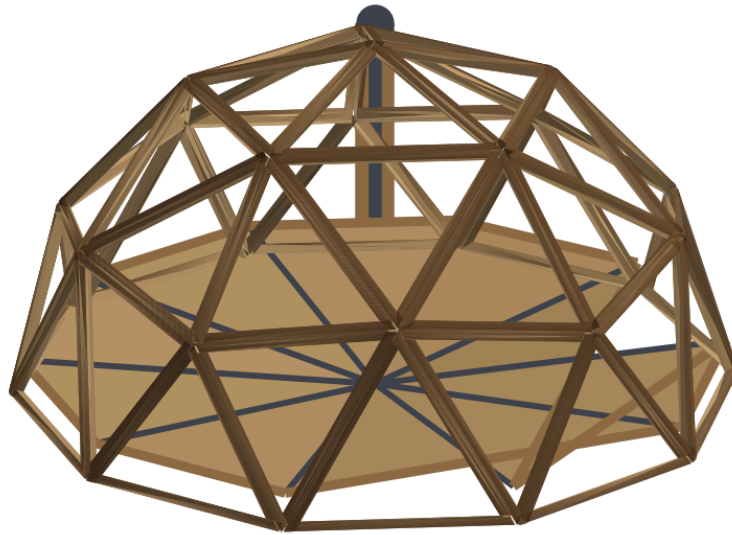
The thing that surprises people

Half a pour is not cheaper than a whole one.

The concrete ring with a timber middle costs *more* than the full slab. The saving on concrete is smaller than the cost of the timber deck that replaces it, and the ring's formwork is fiddlier per cubic yard than a slab's.

This is in the book because the intuition is so strong and so wrong, and because the model prints the number that does not flatter the argument.

The floor that clamps



The mast through the column, and the floor that clamps to it, sitting on the base ring where a floor goes.

There are two ways to build a floor in a dome.

Framed into the base ring. Joists between the ring segments, decking over. Simple, strong, and permanent: it is now part of the structure and removing it is a demolition.

A hub, spokes and decking. A steel hub in the middle, radial spokes out to the base ring, timber over the top. It is a component. It comes out, it goes back, it can be replaced with a better one in ten years, and it can be left behind when the dome moves.

Chapter 3 argues for the second on general principles — the floor is one of the two layers that matters, so keep it changeable. It also turns out to be what makes Chapter 17 possible, which was not the reason for doing it.

The hub wants something in the middle to clamp to. In this design that is the utility column, which is already there, already centred, and already the one thing running floor to apex.

Under-floor is storage

A deck on piers has a void under it. That void is dry, it is out of the way, and it is the cheapest cubic footage in the building.

Design for access to it from inside — a hatch, not a crawl from outside — and it is the difference between a small building that feels cramped and one that does not.

What this chapter will not tell you

Frost depth, soil bearing, seismic anchorage.

Those are local, they are the reason foundations are the part of a building most tightly bound by code, and this book does not know where you are. The five constructions above are priced take-offs of a design; what depth your piers go to is a question for somebody who has stood on your ground.

A Pad, Not a Plot

The ground is the one thing you cannot take with you, so stop buying it

The ground is the one part of this building you cannot take with you.

Everything else — 120 members, forty panels, a column, a cap, a floor — comes apart and moves. The ground does not, and it is usually the most expensive thing in the whole arrangement.

So stop buying it.



The landowner builds the pad. It stays when the dome leaves.

The split

The host owns the ground and everything permanently attached to it. The pad, the service port coming up through it, the water and power to that port, the access track, the under-floor storage if there is any.

The tenant owns the building. The frame, the panels, the shell, the column, the floor, the cap.

That line is not arbitrary. It falls exactly where “can this be carried away” falls, which means it never has to be argued about. Nothing straddles it.

Why it is a clean line

The usual version of this arrangement — renting land and putting something on it — is messy because the something is half-attached. Whose slab is it? Whose septic? Who pays when the deck rots?

Here the pad is a single object with a price. In this model it is \$6,240, built once, by the host, to a specification the tenant can check before arriving. The dome is another object with a price, \$12,036, and it arrives on a trailer.

When the arrangement ends, the tenant takes a building and the host keeps a platform. Both of those are worth something to their owner and neither is worth arguing about.

What the host is really selling

Not land. A **prepared, serviced, level place to put a building**, which is a much more specific thing and much more valuable than an acre of field.

Preparing that place is the expensive, skilled, permitting-heavy part of building anywhere. It involves earth, concrete, a trench, and somebody who knows where the frost line is. Doing it once and having it used for decades is a far better use of that effort than doing it once per owner.

And what the tenant is buying

A foundation, without a mortgage.

The pad is cheaper than the foundation they would otherwise pour, because it is amortised over more than one occupancy. It is available now rather than after a land purchase. And — the part nobody expects to value until they have it — it is *leavable*.

The number that decides it

The pad is \$6,240. The floor a dome brings with it is a few thousand more. Against a piece of land, which in most places is a five-figure number before anything is built on it and a thirty-year commitment after.

That ratio is the entire argument for the next chapter, which works it through from both sides and says what would have to be true for it to hold.

PART 6

The Network

“Own the building. Rent the ground. Move when you want to, and leave a platform behind rather than a house.”

One chapter, the economics of a pad network from both sides of it: what the dome owner saves, what the landowner earns, and what would have to be true for either.

Why a Network Beats a Park

An RV park rents you a slot for a week. This rents you a foundation for a decade, and both sides come out ahead

An RV park rents you a slot for a week. This rents you a foundation for a decade.

That is the whole difference, and everything below follows from it. The arrangement only works if it is better for the landowner *and* for the person who arrives with a building, so this chapter works it out twice.

First, why it is not a caravan site

An RV park sells **nights**. Its costs are turnover: booking, cleaning, hookup, disconnect, a pitch standing empty out of season, and a tenant who is leaving from the moment they arrive and behaves accordingly.

A pad sells **years**. It is built once, connected once, metered once, and occupied by somebody who owns the building standing on it. The revenue per square foot is lower. The cost per occupied month is very much lower. The difference is turnover, and turnover is most of what makes short-stay accommodation a job rather than an income.

The tenant’s own building is also the deposit. Nobody abandons a dome.

The tenant’s side

Four ways to be housed, on this book’s dome and on rates declared in the park model with the assumption written beside each one:

	per month	at the door	over 10 yr	owned after
hotel room	\$4,200	\$0	\$504,000	\$0
short let	\$2,850	\$0	\$342,000	\$0

apartment lease	\$1,450	\$3,625	\$175,450	\$0
own the dome, rent the pad	\$528	\$12,036	\$75,365	\$12,036

Two things in that table are worth slowing down for.

The monthly figure includes the dome wearing out. \$528 a month is the pad's rent plus the dome's own depreciation over a 30-year service life. It is not rent-and-ignore-the-asset; the building is being used up and the table says so.

The entry cost is the real objection. \$12,036 at the door against an apartment's \$3,625. That is the catch and it does not go away. What the table shows is how quickly it is repaid: owning overtakes renting after **11 months**, and after ten years the difference is about \$100,085 — with a building still standing at the end of it that you own.

What the tenant actually gets

No land purchase and no mortgage. The most expensive item in almost every housing decision is removed, not reduced.

A building that moves. Not “portable” in the sense of a caravan — this is a real timber shell — but demountable. It comes apart into panels, a column, a floor and a cap, and it goes onto a trailer. The thing you cannot take is the pad, and the pad is the cheap part.

Flexibility in how long. A season or a decade, decided as you go. There is no twelve-month lease to break because the thing tying you to the place is a rent on a platform, not a claim on a home.

Flexibility in where. This is the part that has no equivalent. If the work moves, or the climate turns, or the neighbours do — you take the house. In every other arrangement on that table, moving means losing either a deposit or a building.

A floor that is better than the pitch. An RV park gives you a slab and a hookup. A pad in this arrangement is a prepared, levelled, serviced platform sized for the building that is going on it, and the dome's own floor clamps to its own column.

The host's side

Three things to do with a piece of ground:

	upfront	a year, net	payback
bare ground, unused	\$0	\$-380	never
one serviced pad	\$14,280	\$4,868	2.9 yr
furnish a unit and let it short	\$26,316	\$26,238	1.0 yr

And here is the number that does not flatter the argument, printed rather than buried: **furnishing a unit and letting it short earns the host more.** More than five times the annual net, and it pays back in about a year against a pad's 2.9 years.

So why would anybody build pads?

Because those two lines are not the same kind of thing

The short-let line is given every night of the year full, which nobody achieves. Halve the occupancy and it halves.

It also requires the host to own a *building* — to build it, insure it, furnish it, clean it three times a month forever, hold a reserve against damage, renovate it every year, and answer the phone when the heating fails at eleven at night. That is not an investment. It is a small business, and the income in the table is the pay for running it.

The pad is an annuity. \$14,280 once, \$4,868 a year net, and the work is reading a meter and sending an invoice. The tenant owns the building, so the tenant fixes the building.

The honest framing is: **a short let earns more per pound of effort at the start and a pad earns more per hour of attention forever.** Which of those you want is a question about your life, not about your land, and this book is not going to pretend the money answers it.

What the host is really selling

Not land. A prepared, serviced, level place to put a building — which is the expensive, skilled, permit-heavy part of building anywhere.

Doing that once and having it used for thirty years by a succession of owners is a far better use of the effort than doing it once per owner. The pad is also worth something while it is empty, which bare ground is not: it is an improvement with a rent attached to it.

Why a network and not a park

One host with one pad is a transaction. The thing that makes it a network is that the pads are **the same**.

A dome built to this book's geometry fits any pad built to this book's specification, which means a tenant can move between them without modifying anything, and a host can take any tenant without knowing in advance which dome is coming. Chapter 14's clean split — the host owns what is attached to the ground, the tenant owns what can be carried — is what makes that interchangeability possible, because nothing straddles the line and so nothing has to be negotiated.

That is the same argument as Chapter 8, applied to land instead of labour: a short, fixed interface is what lets a thing scale.

What would have to be true

This chapter is a design possibility, not a tested arrangement. Four things would have to hold, and none of them is proven:

Zoning would have to permit it. A demountable dwelling on a leased pad is not a category most jurisdictions have. Some will treat it as an RV, some as a permanent dwelling, some as neither. This is the single largest risk and it is entirely local.

The rates would have to survive contact. \$494 a month for a serviced pad is a declared assumption against local RV and storage rates, not a price anybody has charged.

Enough domes would have to exist. A network of pads with nothing to stand on them is a field of concrete. The first hosts carry that risk, which is why the model's test build is three pads and not thirty.

Somebody would have to underwrite the first year. A host who builds a pad on the strength of a book deserves better than hope, and the park model carries a twelve-month manufacturer guarantee for exactly that reason.

State those four plainly and the argument is still a good one. Leave them out and it is a brochure.

PART 7

Variations

“The same frame can wear more than one roof, stand on more than one ground, and hang from nothing at all.”

The full gamut of substitutions with a price on each, and the one variation that is a design possibility rather than a tested build.

Every Other Way of Doing It

Eight frame materials, eight sections, sixteen panels, nine claddings and twelve foundations -- priced on this dome, not in the abstract

This book recommends one way of doing each thing. You are owed the others.

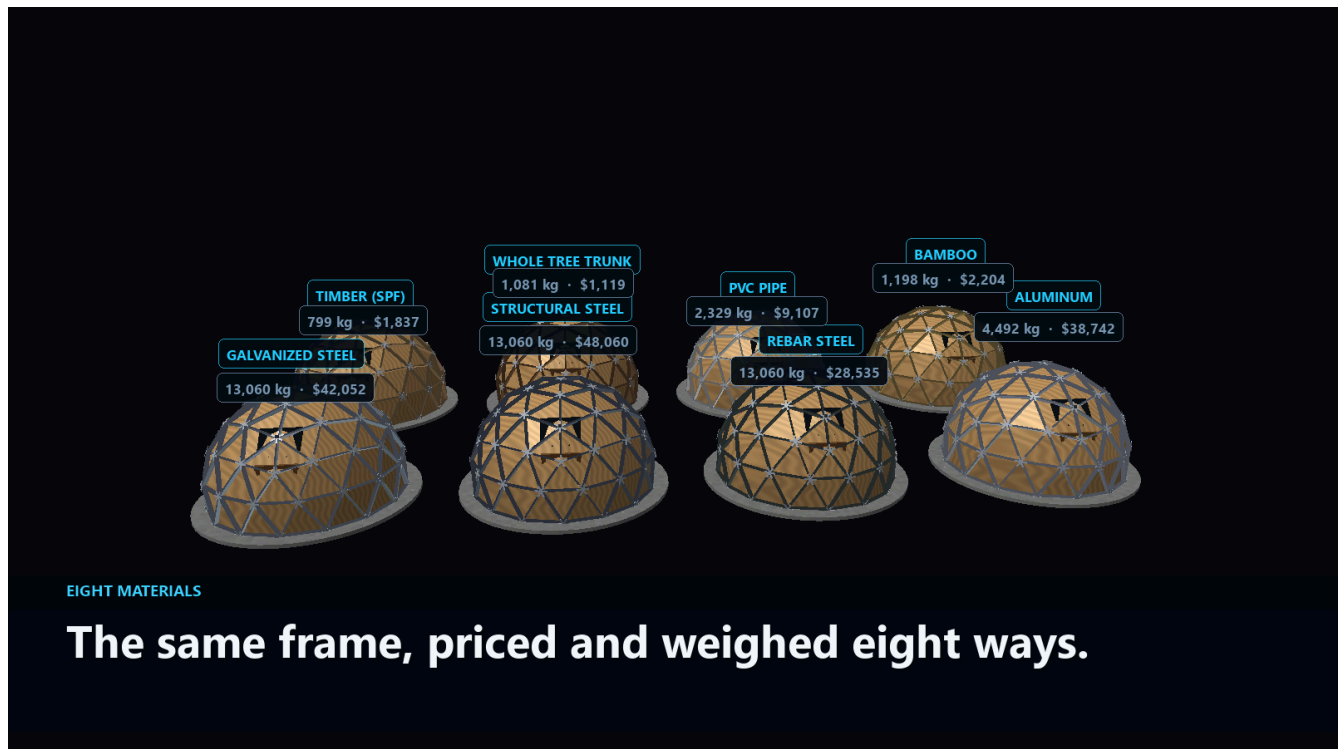
Not a paragraph saying alternatives exist — the alternatives, named, with what each one costs and weighs **on this building**, and a mark against the one the reference build uses.

How to read these tables

A list of materials with prices per square metre is not a decision. A decision is “*on a 19.42-foot dome with 550 square feet of shell, cedar shakes are this much and this heavy, and asphalt is that*” — and that is arithmetic this project can do, because it costs domes for a living.

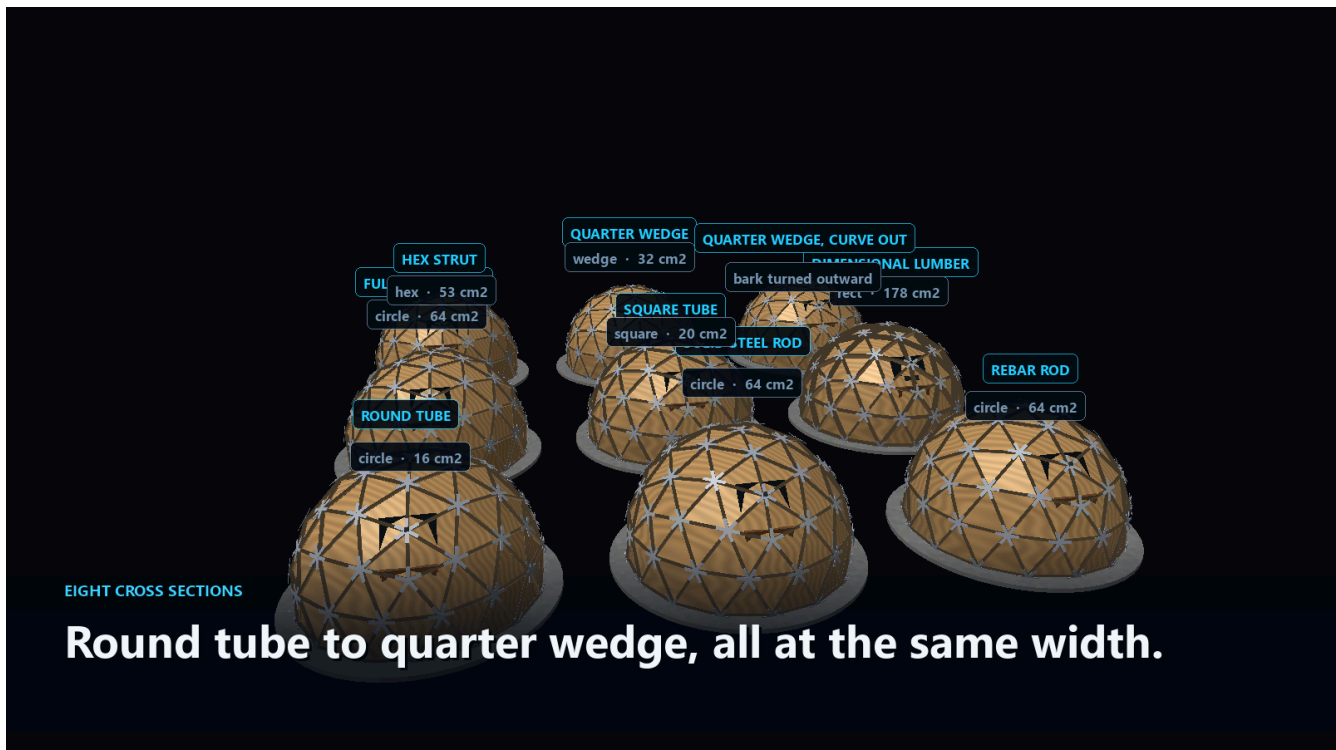
Every option comes from a tool that already models it. Nothing is invented for the book. The full tables are in the reference section at the back; these pages are the shape of the choice.

The frame



The same frame, priced and weighed eight ways.

Eight materials, the same 120 members. The reference build is the whole tree trunk, and it is the cheapest by a long way — which is the entire economic argument of this method in one row.



Round tube to quarter wedge, all at the same width.

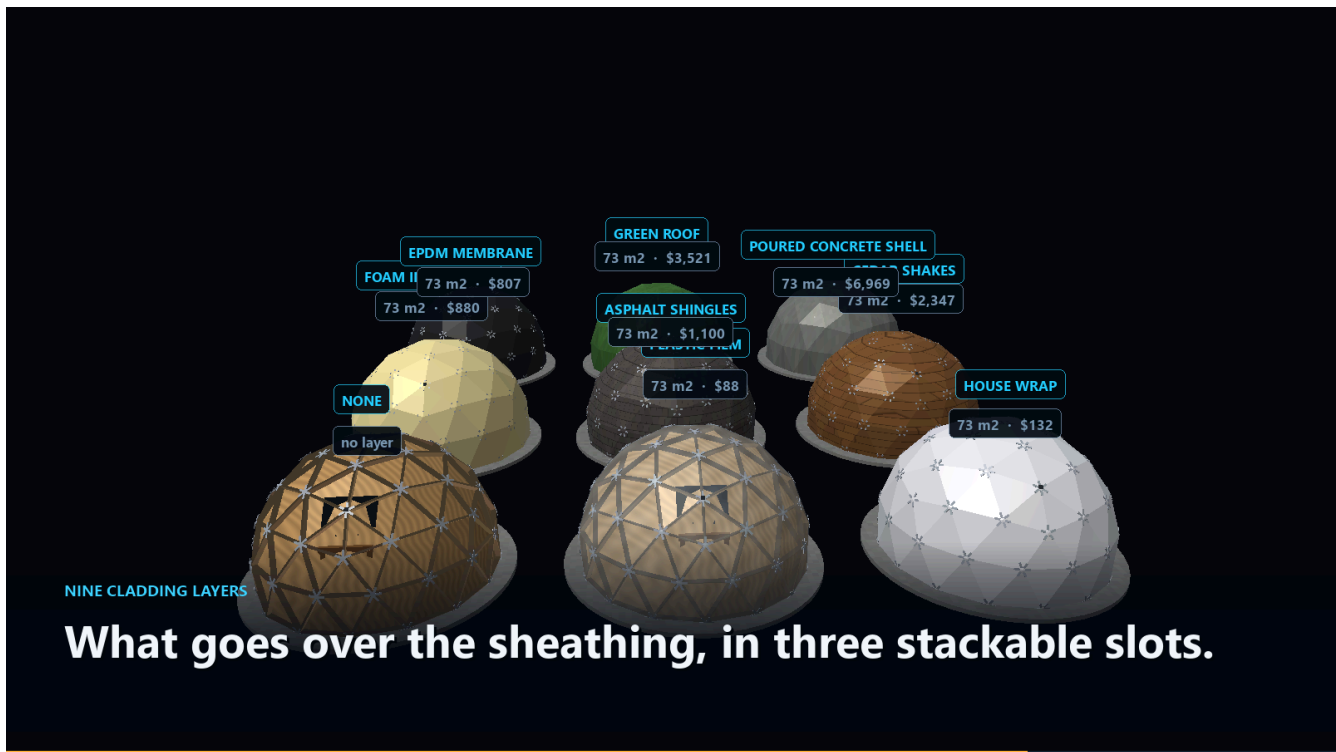
Eight cross sections at the same nominal width. Read the wedge row carefully: the Creator's quarter wedge and this book's member are both split log, sized differently — one is a quarter of a log scaled to the strut's width, the other an eighth of a 12.0-inch one. Same family, different stick.

The envelope



Everything the tool can put between the struts.

Sixteen panel types, from canvas to solar, over 550 square feet. The span from cheapest to dearest is more than a hundred to one, which is why this is the decision that actually sets what your dome costs.



What goes over the sheathing.

Nine claddings. A green roof is twenty-four times the weight of house wrap, and the frame has to carry whichever one you pick.

What is deliberately missing

There is no straw-bale row and no earthbag row.

Not because they are bad — because this project cannot cost one, and a table with a plausible number in it that came from nowhere is worse than a table with a gap.

The same applies to anything else you find missing. If it is not here, it is because nothing in this repository models it, and the honest thing is to say so rather than to estimate.

The four you cannot change later

Everything above can be decided after the dome is standing. Four things cannot, and they are worth a page of their own in the back:

- which way the wedge's point faces

- how many sectors the log splits into
- how big the tree was
- which way the triangle pinwheels

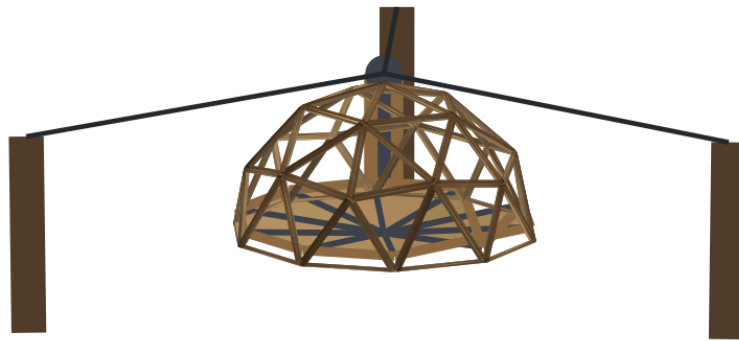
Each of those changes the frame itself, and each has a figure in this book showing what it does to the seam. Chapter 5 is where they are argued; the tables at the back are where they are priced.

Hang It Between Two Trees

The most speculative corner of the method: a mast, three cables, and no ground at all -- priced, and explicitly not rated

This is the most speculative corner of the method, and it exists because the floor of Chapter 13 clamps to a mast rather than being framed into the base ring.

Once the floor hangs off something in the middle, the ground stops being structural. And once the ground stops being structural, you can take it away.



The same mast and floor, hung on three cables to saddles on two trees.

What changes

Everything the earlier chapters assumed about the ground quietly reverses.

The base ring stops being what holds the building up; the mast is. The floor stops being something the dome stands on and becomes something the dome **carries** — clamped to the mast, hanging under the shell, level because the mast is level.

The utility column keeps its job. Services come up a flexible drop instead of a rigid rise, which is a fitting rather than a redesign.

The parts

A steel core through the utility column, timber-clad, floor to apex, with a lifting ring under the seal cap. Three galvanised cables from that ring to three saddles on two trees. A brake winch and a block.

The rig is \$1,490. With the mast and the dome's own floor the whole upgrade is \$4,011, against a frame weighing 2,478 pounds in green pine.

Saddles, not holes. Nothing is drilled into the trees. A saddle spreads the load around the trunk and comes off without leaving a wound, which matters because a tree with a bolt in it is a tree with a problem in ten years.

Why you would

Flood, mostly. Getting a floor above water is the whole reason in some places and no amount of foundation solves it.

After that: slope without earthworks, airflow underneath that keeps timber dry, and the space below, which becomes storage or shade.

The sentence that has to go with it

That is a design possibility, not an engineered structure.

The loads on the trees, the cables, the saddles and the mast need an engineer before anybody stands under it. The model prices the parts; it does not rate them, and the difference between those two things is the whole of somebody's safety.

Trees are also not static. They move in wind, they grow, they shed limbs, and two of them move independently of each other. A rig that is correct on a still day in June is a rig that has not yet been tested.

Nothing in this chapter has been built. It is here because the floor design makes it possible and because a book that only prints the parts of a method that are finished is not documenting a method, it is advertising one.

PART 8

The Tools

“Every number in this book came out of a program about a second before it reached the page. Here are the programs.”

What ships with this book: the solver, the worlds, the cost model and the calculators, with what each is for and how to drive it.

The Software That Wrote This Book

Eight three-dimensional worlds, a cost model and a solver -- what each one is for, what it can do, and how to open it

Not one figure in this book was typed into a sentence.

Every dimension, count, angle, percentage and price is a token in the manuscript that gets filled in, at the moment the book is made, by the same software that draws the three-dimensional models. When this book says the dome is 19.42 feet across, that number came out of a solver about a second before it reached the page.

That software comes with the book. 11 programs — 5 worlds you can fly around in, 4 that are arithmetic with no picture, and the desks where the book and the films are made.

Why a builder would open any of it

The short answer: **because your tree is not my tree.**

Every number in these pages is for one specific dome, cut from one specific size of log, at one split count, with one orientation. Change any of those and every downstream figure changes — the chords, the cuts, the seam key, the timber, the price. The tables at the back are a snapshot of one solve.

The tools are how you take your own.

Three kinds of thing

Worlds are three-dimensional environments with a window, a camera and keys. This is where you go to *look* at something — to turn one decision and watch what it does to the joint.

Models are arithmetic with no picture. They answer a question and print a table: what does this cost, how long is that stick, what does the platform take off to.

Desks are where you make something: this book, the films, the calculators.

What each one is for

WORLDS -- things you fly around in

The raw-wedge solver

You want to see the dome this book is about, and turn one decision at a time to watch what it does to the joint.

start it:

```
py -3.12 geodesic_raw_wedge_dome_dihedral.py
```

- solves the whole 2V frame from one number: the long member
 - four wedge orientations, and the seam changes under each
 - two seam keys, three things to put in the channel
 - explodes the panels apart without moving the geometry, so you can see into a seam
 - switches off the wireframe, the node markers, the debug lines and the sacrificial head stock -- which is what turns a screenshot into an illustration
 - exports a full-size panel drawing as SVG, to print and lay a stick on
 - carries the cost console, so the price is over the geometry rather than in another program
- keys: W A S D walk mouse look E explode the panels apart I the mathematical wireframe on and off
H the head-end stock on and off J the fabrication jig C the cost console

The fabrication jig

You are about to build the fixture and you want to see it assembled, one step at a time, before you cut the board.

start it:

```
py -3.12 geodesic_raw_wedge_dome_dihedral.py   (then J)
```

- twelve steps from a bare bench to a finished panel
 - each step says what it is enforcing, not just what it is
 - the red and green fences that make a rotated stick not fit
 - the two cut planes: the flush-cut fence and the receiving face
 - the sacrificial tail drawn separately from the finished member
 - camera stations at each corner, because the corner is where the method either makes sense or does not
- keys: J show the jig [] step through the twelve V walk the camera round the corners P which panel the jig is set for

The Dome Creator

You want a dome that is not this one: a different frequency, a different material, glass instead of ply, a different foundation -- and the price for each.

start it:

```
py -3.12 dome_creator.py
```

- frequency 1V to 4V, and the truncation
- eight frame materials, from bamboo to structural steel
- eight strut cross sections including the quarter wedge
- sixteen panel types: ply, glass, acrylic, twinwall, SIP, shingle, metal, solar, canvas, mirror, precast
- nine cladding layers over the panels
- seven foundations
- a bill of materials for whatever you just built

keys: F frequency M frame material P cycle the panel under the cursor L cladding layers G foundation B print the bill of materials

The seed world

You want to see the product version: the utility column, the shower cap, the quilted layers and the mast, on the dome this book prices.

start it:

```
py -3.12 -m two_v_demo.app --lesson seed_pitch
```

- the utility column and the gasketed cap at the apex
- the shower cap, and the hats that stack under it
- the mast through the column and the floor clamped to it
- the floating rig: three cables to saddles on two trees
- the composite member, drawn at an exaggerated section and labelled as exaggerated

keys: 1-9 the chapters of the campaign film mouse
 orbit wheel zoom

The pad and the park

You are thinking about the ground: what a pad is, what it costs, and what a network of them looks like.

start it:

```
py -3.12 -m two_v_demo.app --lesson dome_park
```

- a pad built one course at a time
- amber for what the host owns, cyan for what the tenant owns
- several domes on several pads, and the shared services between them
- the rotating pad, which is a design possibility and is marked as one

keys: mouse orbit 1-9 the chapters of the park film

MODELS -- arithmetic, no picture

The geometry

You want the numbers: chords, cuts, angles, areas, counts -- for your dome, not for this one.

start it:

```
py -3.12 -m wedge_book.numbers
```

- two chords and two cut lengths at any diameter
- the bite the pinwheel takes, which does not scale
- face areas, seam runs, member volumes, timber totals
- the whole builder's reference, regenerated

The cost model

You want to know what one costs, line by line, with every assumption named and changeable.

start it:

```
py -3.12 -m seed_model
```

- every line carries its own source constant
- materials, labour, overhead and warranty kept apart
- the maker's markup stated as markup on cost, not margin on price, because those are different numbers
- the pad priced separately, because it is the host's bill
- fit-outs, shells and resins swapped without editing anything

The platform take-off

You are pricing the ground and you want boards and piers counted rather than a rate per square foot.

start it:

```
py -3.12 -m pad_deck
```

- five constructions: gravel, slab, ring, deck, deck with ply
- counted line by line, not rated
- sized to the dome plus a walking margin

The shell model

You are deciding what to put over the frame, and you want the R-value and the price of each layer.

start it:

```
py -3.12 -m soft_shell
```

- one watertight cap, and the quilted layers under it
- each hat a size up from the one beneath
- against a laminated hull, priced four ways
- the concerns it will not hide, printed with the answer

DESKS -- where you make something

The book studio

You want to change this book: its outline, its prose, its figures, or the dome it is written about.

start it:

```
py -3.12 -m wedge_book.build
```

- the outline is code, so reordering is an edit and not a rewrite
- every number is a token resolved at build time
- an audit that fails the build if a figure in the prose is one the code cannot produce
- the pictures render from the solver and from the films

The film exporter

Thirty-seven narrated films are built from the same geometry, and you can re-render any of them with your own numbers in it.

start it:

```
py -3.12 -m two_v_demo.app --lesson wedge
```

- one lesson is one Python module: chapters, narration, scenes
- landscape and phone cuts from one render
- captions, chapter thumbnails and description copy as a set
- a still from any second of any film

The one that matters most

If you open exactly one thing, open the solver, and press the key that explodes the panels apart.

Everything difficult about this method is in the gap between two panels — the key, the angle, why the point faces inward, where the error goes. It is invisible in an assembled dome and obvious the moment you can see into a seam, and no drawing in this book, including the section in Chapter 7, teaches it as quickly as turning it in your hands for a minute.

How the numbers get from there to here

Worth stating once, because it is the unusual part.

The manuscript is Markdown with holes in it. A sentence that needs a figure names the number it wants — `dome.diameter_ft` in double braces — rather than printing one. At build time a resolver fills every one of them from the solver, the geometry and the cost model, and an **unknown token is a hard error** — the book refuses to build rather than printing a blank.

The reason is not showing off. A number typed into a paragraph is correct on the day it is typed and silently wrong from the first time anybody changes anything. Nothing fails. The book just quietly becomes untrue in places nobody remembers.

So the arithmetic in this book is a test suite. It runs, it checks itself against the geometry, and it fails loudly.

It caught four things

This is not theoretical. An audit reads the finished prose and checks every figure in it — every number carrying three or more decimals, and every dollar figure — against what the code can actually produce. It found:

- a set of reference designs that **scaled the pinwheel's bite with the dome**, which is the exact error Chapter 6 spends a page warning about. Every stick of the smallest dome was an inch too long;
- a triangle altitude printed as 52.520 where the solve says 52.516 — in the sentence that tells a reader whether plywood fits;
- four prices from before the shell lost a waterproof layer, still being quoted as current;
- a chapter quoting the cost of a whole upgrade as the cost of two of its three parts, double-counting the third.

Every one of those was written by somebody being careful. That is the point: care is not a method for keeping numbers true across time, and a program that re-checks them on every build is.

Change the tree and rebuild

The whole of it — geometry, prices, figures, this book — regenerates from one set of inputs.

Change the trunk diameter to what you actually felled. Change the split count to what your froe will do. Rebuild. Every chapter that mentions a figure now says your figure, without anybody going looking, and the pictures are redrawn from your solve.

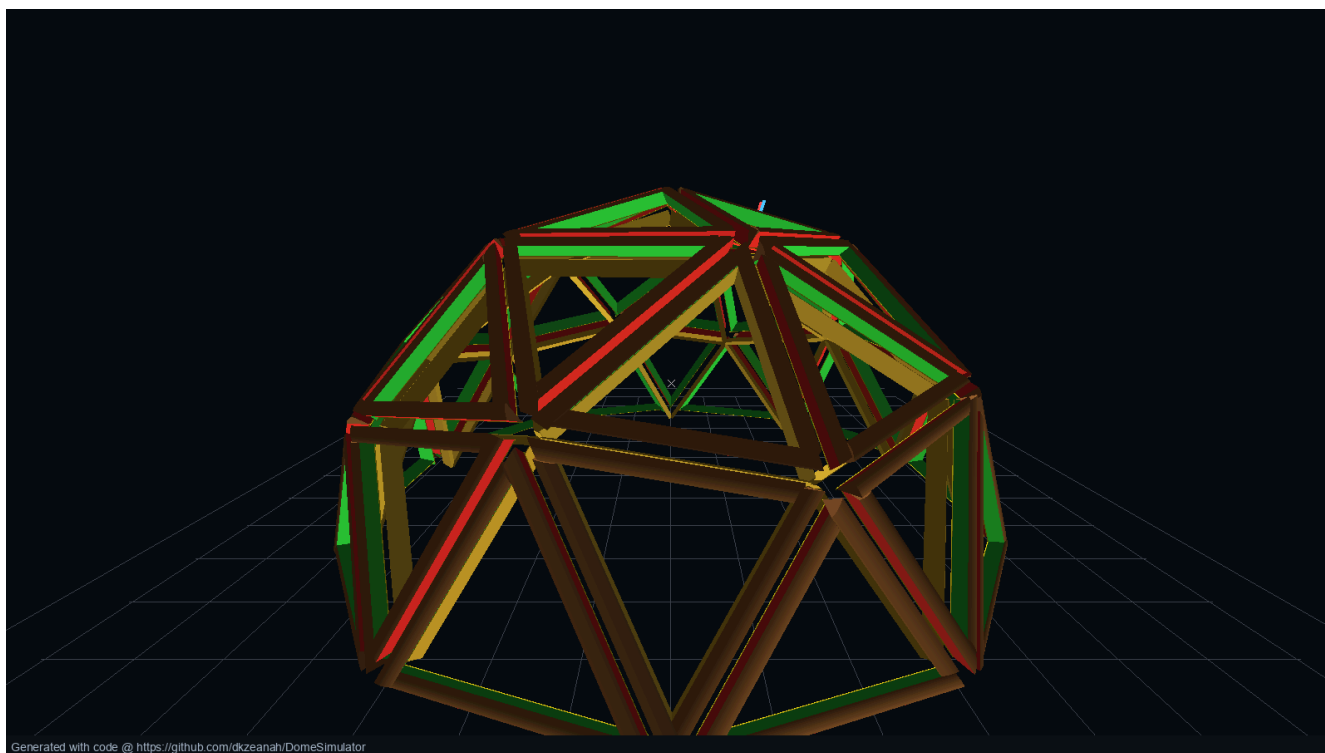
That is what “the software comes with the book” means here. It is not a companion download. It is the thing the book is a printout of.

What Each World Shows You

One picture from every three-dimensional environment in the project, with what it is for and what to do in it

One picture from every three-dimensional environment that ships with this book, what it is for, and what to do once you are in it.

The raw-wedge solver



The reference build, opened eight inches.

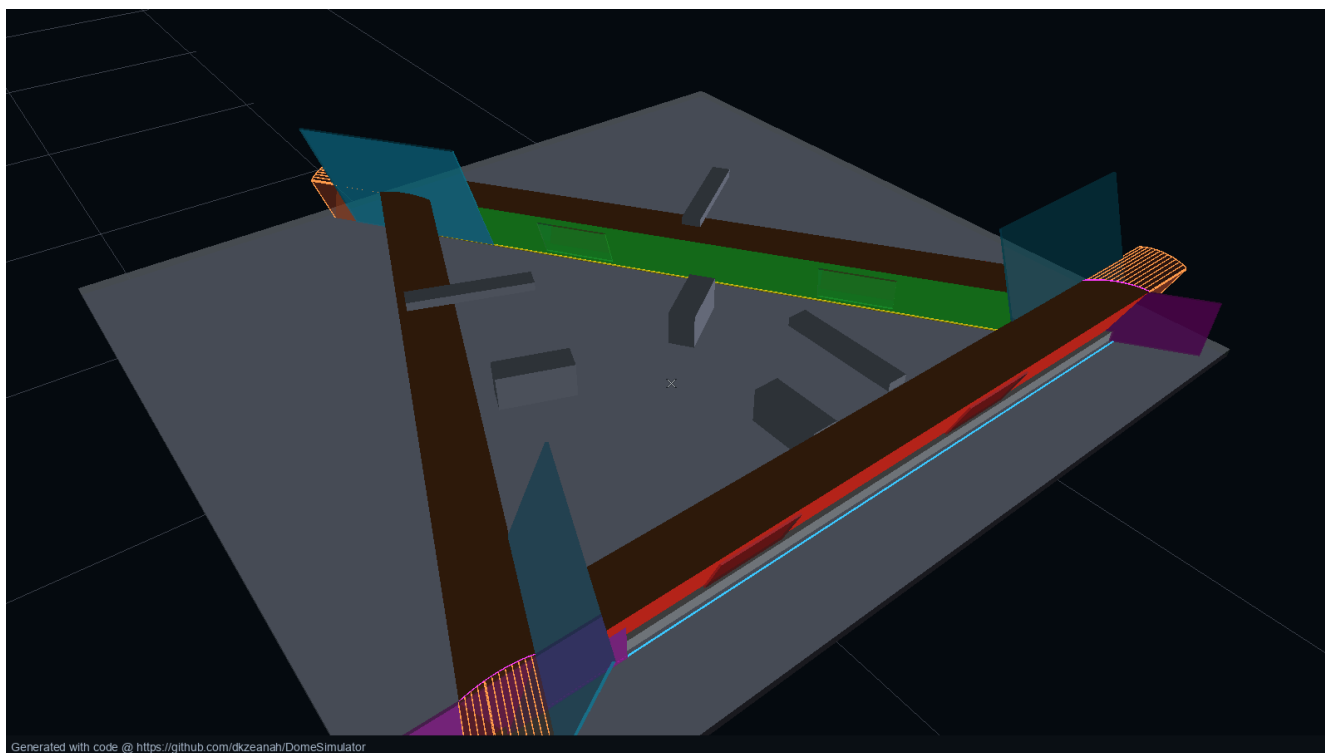
What it is. The dome this whole book is about, solved from one number: the long member. 120 members, 40 panels, 55 seams, every angle derived.

What to do in it. Press the explode key. The panels slide apart along their own face normals — nothing in the geometry moves, it is the same solve drawn with a gap — and for the first time you can see into a seam and watch the key sitting in it.

Then turn one decision at a time. Change the wedge orientation and watch the key's size change. Change the split count from 8 to twelve and watch the seam channel shrink to something that will not take a hose.

What it is not. It has no doorway, no foundation and no utility column, and it builds 2V hemispheres only. Those live in the other worlds.

The fabrication jig



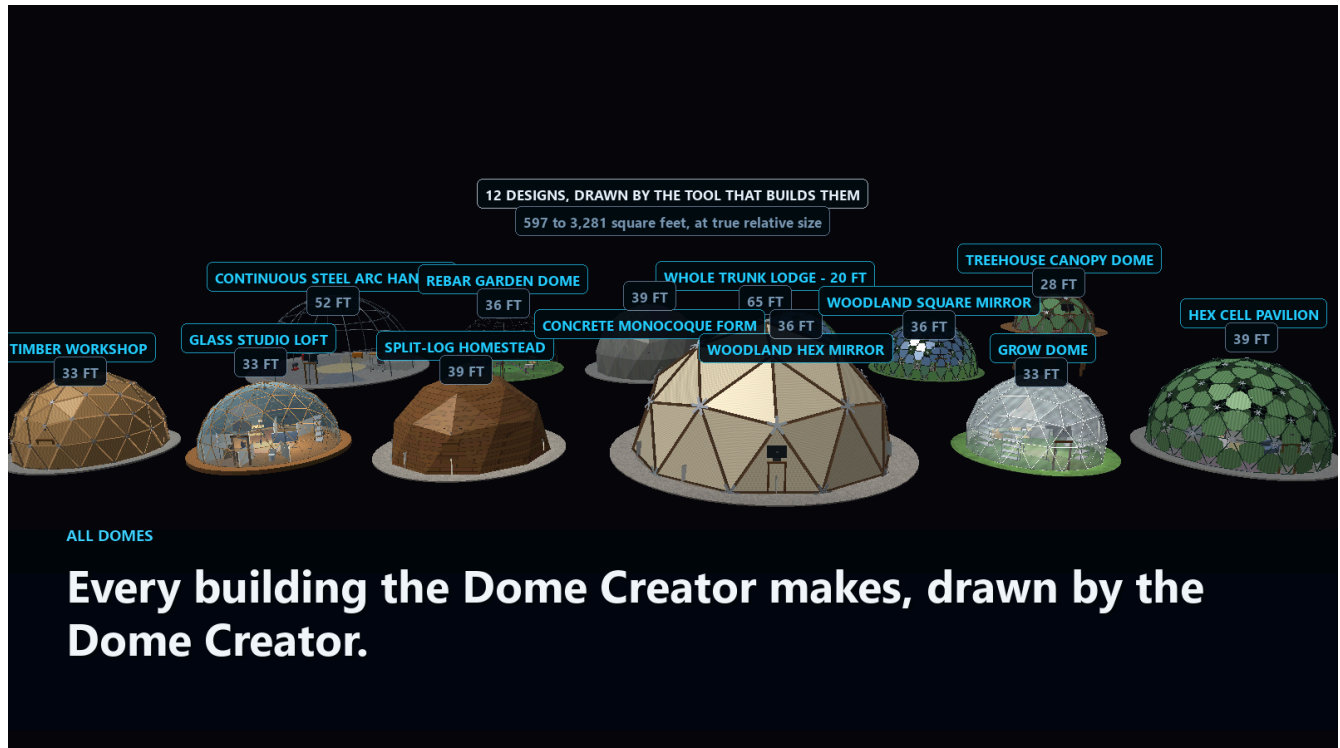
A panel closed on the jig.

What it is. The fixture of Chapter 9, in twelve steps, each one saying what it *enforces* rather than just what it is.

What to do in it. Step through all twelve before you cut a board. The corner stations are the ones worth flying to — the corner is where this method either makes sense or does not, and seeing the butt land on a radial face rather than on an end is the moment it clicks.

Then set it for the other panel family and step through again. The fixture does not change. That is the claim, and you can check it.

The Dome Creator



Twelve designs, drawn by the tool that builds them.

What it is. Every dome this project can describe that is *not* this one. Frequencies 1V to 4V, eight frame materials, eight strut sections, sixteen panel types, nine claddings, seven foundations — and a bill of materials for whatever you just assembled.

What to do in it. Build the dome you were going to build before you read this book, price it, then build this one and compare. That comparison is more persuasive than any argument in these pages, and it is yours rather than mine.

The seed world



Power and water up the middle and out through the apex.

What it is. The product version: the utility column, the gasketed cap at the apex, the shower cap, the quilted layers, the mast and the floor.

What to do in it. Look at the column. It is the one object that touches the ground, the wall and the sky, and almost every decision in Part 5 and Part 7 of this book is downstream of it existing.

Then add layers to the shell and watch the cap grow. Chapter 12 is arithmetic on a page; here it is a thing changing size.

The pad and the park

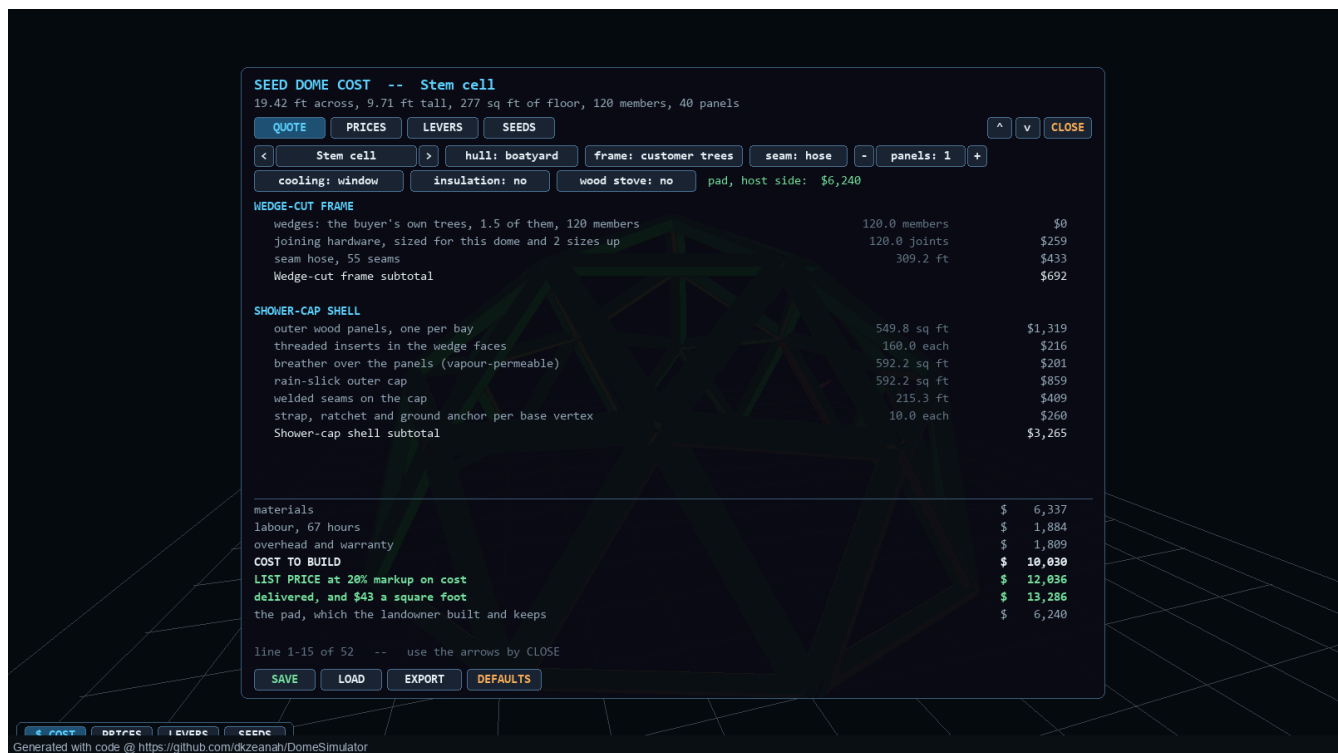


The platform, built one course at a time.

What it is. The ground of Chapter 14: a pad built course by course, and a network of them with the shared services between.

What to do in it. Watch the colours. Amber is what the host owns, cyan is what the tenant owns, and the clean split of Chapter 15 is something you can see rather than something you have to be told.

The cost console



The geometry and the price in one program.

What it is. The cost model, open over the solved dome. Every line carries its own source constant, and changing an assumption moves every figure downstream of it immediately.

What to do in it. Find a price you disagree with — there will be one — and change it. The whole quote re-derives. That is the right way to argue with this book: not by doubting the total but by editing the input you think is wrong and seeing what it was worth.

The number to look at first. \$12,036 for the standard article, built to \$10,030, at \$43.45 a square foot of floor. The maker's markup is stated as a markup on cost rather than a margin on price, because those are different numbers and the one a reader can check with a calculator is the first.

Where to start

If you have an afternoon: the solver, exploded, then the jig's twelve steps.

If you have a tree: the cost console, with your own trunk diameter typed into it.

If you have ground: the park.